

Unit-I

Indian Industrial Environment – competence, Opportunities and Challenges. Entrepreneurship and Economic growth. Small Scale Industry in India, Objectives, Linkage among small, medium and heavy industries. Types and forms of enterprises.

Definition of an Entrepreneur:

Entrepreneur is a person who isolates purchase and sales Opportunities and exercises his intelligence and overcoming spirit to utilise such opportunities to increase his economic power.

Entrepreneur provides the enterprise with resources.

According to Noah Webster, Entrepreneur is “One who assumes the risk and management of business”. It does not involve wages, rent or interest, routine human efforts (i.e., labor), use of land, provision of capital.

According to Higgins ‘Entrepreneurship is meant the junctions of seeking investment and production opportunities’.

“One who can manage a business, one who can organize people for ongoing work”.

Entrepreneur is one who innovates, raises money, assembles inputs, chooses managers, and sets the organization going with his ability to identify them and opportunities which others are not able to identify, and is able to fulfill such economic opportunities.

- Joseph A Schumpeter

The true entrepreneur is one who is endowed with more than average capacities in the task of organizing and coordinating the various factors of production.

- Francis A Walker

Entrepreneur is one who always searches for change, responding to it and exploits it as an opportunity.

Entrepreneurs have to learn to practice systematic innovation which consists in the purposeful and organized search for changes and in the systematic analysis of the opportunities such changes might offer scope for economic or social innovation.

- Peter Drucker

Entrepreneurs and entrepreneurship

Concept:

All human beings have an urge to exercise power over other beings and objects.

Degree may vary person to person.

Entrepreneur is a person who isolates purchase and sales opportunities and exercises his intelligence and overcoming spirit to utilize such opportunities to increase his economic power. He is a person who assumes the risk and management of his business.

The representative-ness of human agencies which strive for industrial development and are responsive to the business incentives and motivation, is termed as **entrepreneurship**.

Entrepreneur is the person who provides the Fourth factor of production namely **enterprise**. The three factors are Labor, land and capital. The fifth factor is Science and technology.

Classification of entrepreneurs:

1. Innovative entrepreneurs
 - Aggressive in experimentation
 - Sees the opportunities for introducing new things
 - Clever in putting attracting possibilities in to practice
2. Adoptive or imitating entrepreneurs
 - Ready to adopt successful innovations initiated by others
 - Imitates techniques and technology
 - Much suitable to developing countries
3. Fabian entrepreneurs
 - Show great caution and skepticism in practicing any change
 - Have neither will to introduce new changes nor desire to adopt new methods
 - Shy and lazy
 - Try to follow the footsteps of their predecessors
 - Follow customs, tradition and past practices
4. Drone entrepreneurs
 - Refusal to adopt and use opportunities to make changes in production
 - They may suffer losses but do not change production methods
 - They are likely to be pushed out of market because of their uneconomical operations
5. Inheritance
6. Technologist
7. Forced

Other classification may include:

- | | | |
|---------------------|---|-------------------|
| 1. Individual | - | Institutionalized |
| 2. First generation | | Established |
| 3. Men | | Women |
| 4. Rural | | Urban |
| 5. Born | | Manmade |
| 6. Small scale | | Large scale |

Functions of an entrepreneur:-

- 1) Innovation
- 2) Risk-taking
- 3) Organization and management of business so as to have leadership and control over it.

In developing or in a developed country, the state has to assume the role of protection and promoter of small enterprises by way of

1. Allotment of land/factory sheds at low cost by the state
2. concessional finance
3. tax exemptions
4. priority in government purchases
5. Exclusive reservation of certain products for manufacturing.

Entrepreneur distinguished from other functionaries:-

1. Inventor loses to entrepreneur in the sense that inventor invents but cannot exploit economically.
2. Entrepreneur and Speculators - Resembles but differs in organising and introducing innovations.
3. Entrepreneur and inventor: - Entrepreneur exploits the inventions commercially into a business.
4. Entrepreneur and Manager

Entrepreneur and other factors of production such as rent of land, wages of labour and interest for capital will always be positive, but the reward of the entrepreneur are – profit – can sometimes be negative depending upon the degree/grade of entrepreneurial talents.

	Entrepreneur (Self-employed)	Manager (Paid employee)
1	You are completely independent and free to work according to your wishes.	You have to work according to the directions of the employer.
2	You are entitled for the whole profit you make. Your efforts are rewarded proportionately.	You get your fixed salary or at the most a small amount of bonus as reward for your work.
3	There is great scope of creativity. You can introduce innovation and changes to improve your earnings.	There is always lack of incentives so there is generally lack of enthusiasm.
4	As your efforts are quickly rewarded you feel encouraged to work with enthusiasm.	There is little or almost no scope for creativity. Rather the work is generally of a monotonous nature.
5	In self-employment there is full responsibility. Whatever happens has to be borne by the self-employed person.	Comparatively there is less responsibility.
6	There are risks of less income or loss at times.	There is no risk of less income as any paid employee is paid fixed salary.
7	In many cases self-employed persons are not qualified or even illiterate	For paid-employment, generally one has to be qualified or skilled in one or the other trade.
8	Initiative, capital and hard work is needed to make self-employment a success.	There is no need of money and once appointed, there is no compulsion to take initiative or work very hard.
9	A self-employed person gives jobs to other people and thus helps the nation in solving the unemployment problem.	It is difficult to get paid employment and there is no scope of his giving employment to others.

Qualities of Entrepreneur:—

1. Capacity to assume risks and possessing self confidence.
2. Technological knowledge, alertness to new opportunities, willingness to accept change and ability to imitate
3. Ability to marshal resources — ability to build organisation is perhaps the most precious of all entrepreneurial skills.
4. Ability to organisation and administration.

The entrepreneur must have special quality of judgment, perseverance and knowledge of the world as well as of business. He is called upon to estimate, with tolerable accuracy the importance of

specific product, the probable amount of demand and the means of its production; at another, buy or order the raw material, collect labourers, find consumer and give at all times a rigid attention to order and economy.

Definition of Business:—

Business means the state of being busy. It is a gainful human activity that one can be busy about. It is an elastic composite word covering the complex gainful activities. The principal purpose is to extract, transform and exchange the natural resources with the object of selling them at a profit.

Business includes trade, commerce and industry.

Trade — domestic or internal trade,
foreign trade or international trade

Commerce involves the distribution process consisting of buying and selling (trade) and all other activities of storing, grading, packing, financing, insurance and transporting to places of requirements. Principal function of commerce is to remove hindrances of person, place and time in connection with the distribution of commodities until they reach the customers.

Industry involves in process of extraction and production of goods which are sold either for further transformation into finished product or for ultimate consumption. The wholesome knowledge of business system will be making an individual a successful entrepreneur.

Entrepreneur word is derived from French word *entreprendre* meaning "to undertake".

The "Entrepreneur" means a person who organises and manages a business undertaking assuming the risk for the sake of fulfillment of some of his needs.

The Objectives of Business or an industry:—

1. Economics 2. Social 3. Human

1. Economics:—

Profit making is one of the objectives.

Profits are the legitimate rewards for honest endeavors. Profits are also the key to efficiency, drive and initiative. Profits also serve:

- a) Providing capital for meeting the cost of enterprise and to continue the same,
- b) Provide capital for future expansion,
- c) Provide a measure for business performance,
- d) Provide an indicator of efficiency.

2. Social:—

Business exists in society. Hence the survival of the industry/business rests in its dealing with society. Society provides customers. The requirements are right quality of goods in right time and at right price and these are conflicting to the view points of business/industry.

— By way of creating welfare activities, hospitals, schools, entertainment etc.

3. Human:—

Business or industry is an organised system of human beings for joint performance to meet the social needs. Building up a high team spirit and healthy atmosphere is the essential function of an industry, to realise maximum of talents and potentials of workers or employees.

WORK: - 1) economic in nature and 2) motivational

Types of work are

- 1. Extractive work
- 2. Constructive work (production)

3. Distributive work (trade)
4. Professional work
5. Defense and allied services

Some people work for others while some like to work for themselves. In other words some people are employed by others, some are self-employed (Entrepreneur).

Characteristics of an enterprise:

- 1 People work together primarily for the purpose of making, selling and / or distributing a product or service
- 2 It consumes other valuable resources to produce the output.
- 3 It is a continuing entity.
- 4 It must have enough autonomy for its management to take actions considered appropriate to maintenance or enhance the firm's economic success.

SMALL SCALE UNIT: - A unit where investment in fixed assets in plant and machinery is only up to Rs. 5 Crores (in case of ancillary units the limit is up to Rs. 5 Crores and 25 Lakhs) is defined as a small scale unit, whether held on ownership terms or by lease or by hire-purchase. These should not be subsidiaries of large units.

ANCILLARY UNIT: - Small scale industries producing items for large industries needed costly machinery and equipment. These industries were producing standardized goods of high precision to suit the requirements of big and reputed large scale units and installation of costly testing equipments was also necessary, at least 50% of the annual production is meant for a large scale industry, is termed as Ancillary industry. A unit which produces parts, components sub-assemblies for supply against known or anticipated demand of one or more large units manufacturing or assembling complete products and which not a subsidiary to or controlled by any large unit in regard to the negotiation of contracts for supply of its goods to any large unit.

Large scale industries: - Earlier all those industries which require Letter of Intent (LOI) and industrial license are termed as large scale industries. These industries had to register with Directorate General of Technical Development (DGTD). As per the latest Government definition, all industries with investment in plant and machinery more than RS. 100 crores come under large scale industries.

TINY UNIT: - Units with investment in machinery and equipment up to Rs. 25 Lakhs and situated in towns which have a population below 50 thousand is defined as a tiny unit.

Self-employment opportunities are available for tiny units, for ex-servicemen, technicians etc. Annual turn over of tiny unit will be less than Rs. 50 lakhs. Prices of the goods manufactured by a tiny unit are fixed higher than that of a small scale unit.

Export Oriented Units (EOU):-

Investment up to Rs.10 crores. Exporting a minimum of 30% of its production within 3 years of commencement of production.

Manufacturing entrepreneurship did not develop till 1850 due to lack of communication and transportation systems. Banares, Allahabad, Gaya, Puri which are on the riverbeds did business.

Small Scale Industries in India:

- The role of SSI units is significant in overall growth of the economy in our country.
- SSIs are especially important in the context to employment potentials, equitable distribution of wealth, balanced regional growth and preservation and development of ancient art and craft.

- At present in our country the small scale sector contribute nearly 55% of the total industrial output and 40% of the total exports.
- SSI has also achieved a high degree of sophistication and has been making significant progress in quality up gradation and standardization.
- Small enterprises are able to successfully adapt to the changing situations and possess creative strength.
- SSI has low capital intensity and high labor orientation.
- Although the prospects of SSI are plenty, the new industrial policy is hampering their sustained growth in the nineties. The new policy is more favorable to the multinational companies.
- The rapid and diversified growth of small-scale units has been contributing to the nation's economic development.

Objectives:

1. Expanding employment opportunities
2. Adoption of modern techniques
3. Dispersal of industries in rural areas
4. Production of consumer goods on large scale
5. Mobilization of local skills and capital
6. Equalitarian distribution of income and wealth widely in society
7. Bring out the latent potentials and skills in the entrepreneurs

Linkage among small, medium and heavy industries:

The relation ship between large scale, medium scale and small scale industries should be mutually beneficial and helping each other rather than competing and conflicting.

Large scale industries undertake the manufacture of much bigger and complex machines or systems which demand high technology involving large number of components and sub-systems. Often it may not be economical and possible to manufacture every item with in the organization. These items can be profitably outsourced to either medium or small scale industries. Thus many SSI or ancillary industries can be developed to meet their production requirements.

As the overheads of small scale industries are nominal, the cost of production will be much lower compared to the cost of production in the large scale units.

With the guidance and support rendered by the big industries, the quality of the products can be improved to the levels required.

If a number of sources are developed for a product, it is possible to increase the quantity of output as required and uninterrupted supplies can be maintained.

For the over all economic growth of a nation, the development of all the three sectors is equally essential.

Various reasons for lack of development are:

1. Lack of capital
2. Lack of political unity
3. Network of custom barriers
4. Existence of innumerable systems of currency
5. Regional markets plagued by arbitrary political authority
6. Taxation policies
7. Low prestige of businessmen in the society.

Agency Houses introduced new methods of production, new sources of raw materials and

new products and markets.

These houses entered business, trade, banking in Calcutta, Bombay and Madras. Parsees and Marwaries dominated the business and industry from 1st world war period and 2nd world war period.

Indian Industrial Environment

1.0 Immediately after our country became independent, Government of India felt the need for developing industries in a planned manner in the national interest. The government adopted an industrial policy 1956 with the objectives of

1. To Accelerate rate of economic growth through speedy industrialization
2. To develop heavy industries and machine building industries
3. To expand the public sector
4. To build up a large and growing cooperative sector
5. To reduce disparities in income and wealth
6. To prevent private monopolies

Main features:

1. Classification of industries in to Schedule A, B and C
2. Sectoral interdependence
3. Role of small-scale and cottage industries
4. Reduction of regional disparities
5. Assistance to private sector
6. Sound industrial relations and incentives for labor
7. Management of public undertakings

Due to the changing conditions, amendments to the industrial policies have been made during 1973, 1977, 1980, 1991, until the recent reforms of 1992-93.

In the Indian business scene, many changes have been introduced in the liberalization process of economy which began in 1990 and still continues.

The major changes are:

1. Industrial policy changes
 - Very few industries requiring licenses
 - Liberalizations in FERA
 - Almost no control under MRTP act
 - Role of PSUs getting diluted
 - Liberalizations of foreign direct investment
 - More freedom in capital market- abolition of CCI
2. International trade policy changes
 - Globalization of economy
 - Lowering down of import tariffs
 - More items under OGL
 - Emphasis on export but not through financial incentives
3. Structural adjustment
 - Phasing out of subsidies
 - Gradual dismantling of administrative price mechanism
 - Public sector disinvestment

- Exit policy for business
- Smother way of merger and acquisition

1.1 The present industrial environment in India:

Environment can be broadly classified into the following categories

1. Economic environment

- Economic system
- National income and its distribution
- Monetary policy
- Fiscal policy
- Natural resources
- Infrastructural facilities
- Raw materials and supplies
- Plant and equipment
- Financial facilities
- Man power and productivity

2. Political-legal environment

- Political stability
- Provision various incentives and support
- Rules and regulations
- Defense expenditure

3. Technological environment

- Technology developed within India
- Import of technology
- Research & Development facilities

4. Socio-cultural environment

- Expectations of society from the business
- Attitude towards business
- Customs , Traditions and conventions
- Level of education
- Life style of people

5. Competitive environment

- Industry setting
- Industry attractiveness
- Industry structure

1.2 Competence:

- Investment capabilities are enhanced to a great extent
- Infrastructural facilities like land, buildings, power, fuel, transport, communication network etc. are vastly improved
- Technology can be obtained easily. Various R & D laboratories are available for developing process and products
- Skilled labor, technical experts and professional managerial skills are available.

- Capability of meeting international quality standards
- Governmental support and encouragement

1.3 Opportunities:

- A number of institutions and organizations are engaged in the development of entrepreneurship.
District Industries centre (DIC)
Indian investment centre (IIC)
Small Industries Development Organization (SIDO)
Small Industries Service Institutes (SISI)
National research Development Corporation of India (NRDC)
National Small Industries Corporation Ltd. (NSIC)
National Alliance of Young Entrepreneurs (NAYE)
Technical Consultancy Organizations (TCO)
APITCO, KITCO etc.
- A number of financial institutions are assisting the entrepreneurs in meeting their capital requirements.
State Financial Corporations (SFCs)
IDBI, IFCI, ICICI, etc.
- Availability of highly skilled labor, Engineers, managerial experts
- Large population creating vast and diversified market
- Government support and encouragement by way of incentives and subsidies.
- Availability of research and development laboratories both in public and private sector.
CSIR, CITD, CIPET, ETDC, ERTL etc.
- Availability of Communication network and Information technology
- Availability of land, Industrial estates and infrastructural facilities

1.4 Challenges:

- Competing in the global market both in terms of quality and cost of production
- Concentration on promising areas ensuring high quality, attractive packaging, acceptable delivery, adequacy of supplies, strong home market
- Increasing the capacity of borrowing on commercial terms. (Increasing credit worthiness)
- Emphasis on knowledge management
- Acquiring high technology and improved processes and constant updating

2.0 Entrepreneurship and Economic growth:

Entrepreneurship is a creative activity or an innovative function and can be seen in the form of

1. Introduction of a new product
2. Use of new method of production
3. Opening a new market

4. Conquest of new source of supply of raw materials
5. New form of organization

All these factors are contributive to the economic growth. Entrepreneur acts as an agent of production bringing together all factors of production and providing management and control for the survival and growth of the production unit.

Economic development consists of employing resources in a different way, bringing in a new combination of means of production. The entrepreneur looks for ideas and put them into effect for economic development.

Economic growth depends on the rate of innovation and the rate of technical progress in the economic field.

Entrepreneurs, with high level of achievement, accelerate the process of economic development.

Entrepreneurship is a process of creating wealth by bringing together the resources in a new way to start a venture that benefits the customers and rewards the founders.

All the developed nations have been benefited from the entrepreneurs in building the economy.

Entrepreneurship leads to industrialization which has better linkages to the other sectors of economy like generation of electricity, exploitation of natural resources, network of transport and communication systems, and expansion of markets.

Industrial development provides vast employment opportunities.

It gives the advantage in international trade.

It is an effective instrument to the growth and welfare.

Industry being an important component of national income, augments not only the taxable capacity of the people in the country, but also contributes effectively to the revenue by paying more and more direct and indirect taxes which can be used by the government for the welfare of its citizens.

Thus the entrepreneurship plays a very important role in accelerating the economic growth.

Types and Forms of enterprises

1.0 Types of enterprises:

1. Service enterprises

- Consulting agencies
- Repair shops
- Professional – doctors, lawyers, engineers, chartered accountants
- Beauty parlors
- Carpenters

2. Trading enterprises

- Provisions shop
- Medical shops
- Dealership and agencies
- Clothes merchants

3. Manufacturing enterprises

2.0 Forms of enterprises:

1. Sole proprietorship
2. Partnership
3. Joint stock companies
 - Public limited
 - Private limited
4. Co-operative enterprises

1. Sole Proprietorship:

Definitions:

It is the form of business organization on the head of which stands an individual as the one who is responsible, who directs its operations, who alone bears the risk of failure.

A sole trader is a person who sets up the business with his own resources, manages the business himself employing persons for his help and alone bears all the gains and risks of the business.

A sole trader is a person who carries on the business exclusively by and for himself. He is the supreme judge of all matters pertaining to his business subject only to the general laws of the land.

Characteristics of Sole Proprietorship:

- | | |
|----------------------------|--|
| 1. Individual initiative | 5. Secrecy |
| 2. Unlimited liability | 6. Proprietor and proprietorship are one |
| 3. Full managerial control | 7. Owner and business exist together |
| 4. Motivation | 8. Limited area of operations |

Advantages:

- | | |
|----------------------------------|--------------------------------------|
| 1. Easy formation | 8. Direct accessibility to consumers |
| 2. Better control | 9. Inexpensive management |
| 3. Flexibility in operations | 10. No legal restrictions |
| 4. Retention of business secrets | 11. Socially desirable |
| 5. Easy to raise finance | 12. Self employment |
| 6. Direct motivation | 13. Good relations with employees |
| 7. Promptness in decision making | 14. Benefit of inherited good will |

Disadvantages:

- | | |
|-------------------------------|--------------------------------|
| 1. Limited resources | 5. Limited scope for employees |
| 2. Limited managerial ability | 6. No large scale economies |
| 3. Unlimited liability | 7. More risk involved |
| 4. Uncertain continuity | |

Suitability:

- | | |
|------------------------------|-------------------------|
| 1. Small scale of operations | 4. Speculative business |
| 2. Local market | |

3. When personal contact with customers is needed

Examples: Retail traders, Small Engg. Firms, Repair shops

2 Partnership:

Definitions:

The relation between persons who have agreed to share profits of a business carried on by all or any of them acting for all. - As per Partnership act 1932(sec.4)

A partnership firm is a group of two or more persons who have joined capital or services for the prosecution of some enterprise.

Characteristics:

- | | |
|---------------------------------------|--------------------------------------|
| 1. Association of two or more persons | 8. Utmost good faith |
| 2. Contractual relation | 9. Restriction of transfer of shares |
| 3. Earning of profits | 10. Common management |
| 4. Existence of business | 11. Partners and partnership are one |
| 5. Implied authority | 12. Capital contribution |
| 6. Unlimited liability | 13. Protection of minority interests |
| 7. Principle and agent relationship | 14. Continuity |

Advantages:

- | | |
|---|-------------------------------------|
| 1. Easy to form | 9. Close supervision |
| 2. Moderately large resources | 10. Flexibility of operations |
| 3. Greater managerial talent | 11. Secrecy |
| 4. More credit standing | 12. Protection of minority interest |
| 5. Promptness in decision making | 13. Easy dissolution |
| 6. Sharing of risk | 14. Democratic administration |
| 7. Relationship between work and reward | 15. Less managerial expenses |
| 8. More possibility of growth and expansion | |

Disadvantages:

- | | |
|-------------------------------------|-------------------------------------|
| 1. Unlimited Liability | 6. Burdon of implied authority |
| 2. Limited resources | 7. Lack of public faith |
| 3. Instability | 8. Lack of speed in decision making |
| 4. Mutual distrust | 9. Cautious approach |
| 5. Limitation on transfer of shares | |

Suitability:

1. Small and medium scale operations
2. Higher managerial skill and capital requirement
3. Direct contact with consumers

3. Joint stock company:

Definitions:

A joint stock company is a voluntary association of individuals for profit having a capital divided into transferable shares, the ownership of which is the condition of membership.

A company is an association of many persons who contribute money or money's worth to a common stock and employs it in some trade or business and who share the profit or loss, as the case may be, arising there from.

A company means a company formed and registered under the act.

– Sec.3 of Indian companies Act-1956

Characteristics:

- | | |
|------------------------------|---|
| 1. Association of persons | 6. Separation of ownership and management |
| 2. Independent legal Entity | 7. Perpetual Existence |
| 3. Limited liability | 8. Corporate finance |
| 4. Common Seal | 9. Centralized and delegated management |
| 5. Transferability of shares | 10. Publication of Accounts |

Merits:

- | | |
|--|--|
| 1. Accumulation of Resources | 6. Transferability of shares |
| 2. Limited liability | 7. Ability to cope up with changing business environment |
| 3. Continuity of existence | 8. Diffused Risk |
| 4. Efficient management | 9. Democratic set up |
| 5. Economics of large scale production | 10. Social benefits |

Demerits:

- | | |
|---|------------------------------------|
| 1. Difficulty in formation | 6. Evils of factory system |
| 2. Separation of ownership & management | 7. Speculation in shares |
| 3. Fraudulent management | 8. Concentration in economic power |
| 4. Lack of secrecy | 9. Excessive State regulations |
| 5. Delay in decision making | |

4. Cooperative Societies:

Definitions:

The cooperative form of organization is a democratic set up run by its members for serving the interests of themselves.

- Self help through mutual help. All for each and each for all.

It is a society which has its objectives the promotion of economic interests of its members in accordance with the Cooperative Principles.

- Sec. 4 of The Indian cooperative societies Act- 1912

Principles of Cooperatives:

- | | |
|-------------------------------------|--------------------------------------|
| 1. Voluntary membership | 6. Service motive |
| 2. Political & Religious Neutrality | 7. Distribution of surplus |
| 3. Democratic management | 8. Limited interest on investment |
| 4. One man One vote | 9. State Control |
| 5. Cash trading | 10. Cooperative Education & Training |

Types:

- | | |
|----------------------------|----------------------------------|
| 1. Consumer's cooperatives | 4. Housing cooperatives |
| 2. Producer's cooperatives | 5. Credit cooperatives |
| 3. Marketing cooperatives | 6. Cooperative farming societies |

Advantages:

- | | |
|-------------------------------------|------------------------------|
| 1. Open membership | 5. Low managerial cost |
| 2. Service motto | 6. Surplus shared by members |
| 3. Supply of goods at cheaper rates | 7. Check on other businesses |
| 4. Democratic management | |

Limitations:

- | | |
|--------------------------------|---------------------------|
| 1. Lack of capital | 3. Cash trading |
| 2. Lack of unity among members | 4. Political interference |

Unit-II

Identification and characteristics of entrepreneurs. Emergence of First generation entrepreneurs, environmental influence and women entrepreneurs. Conception and evaluation of ideas and their sources. Choice of Technology – Collaborative interaction for Technology development.

Development in any walk has depended to some degree or other on individual qualities of entrepreneurship.

Small enterprises are usually initiated, managed and developed by one or few individuals of these qualities of entrepreneurship. This constitutes one of the major resources in the promotion of small enterprises.

Identifying entrepreneurs has led to both plain and scientific regarding the characteristics of potential entrepreneurs.

Earlier conception of entrepreneur was described as a man engaged in military expedition.

Characteristics of 1st generation of Entrepreneurs:-

(The Entrepreneurs who are entering for the first time into the business world are called 1st generation Entrepreneurs)

1. High need for achievement
2. High need for power
3. Independence
4. Propensity to take risk
5. Personal modernity
6. Sense of efficacy
7. Support
8. Business experience
9. Leadership
10. Mental ability - intelligence and creativity
11. Clear objectives
12. Maintaining business secrecy
13. Human relations ability
14. Communications ability
15. Technical knowledge
16. Emotional stability
17. Self confidence
18. Motivating skills
19. Long term involvement
20. High energy level

Identification of potential entrepreneurs:

Based on the belief that potential entrepreneurs can be identified and trained, financial institutions in Gujarat have established the Entrepreneurship Development Program (EDP) to promote small enterprises by tapping this latent talent.

Effective psychological-behavior testing can identify potential entrepreneurs and careful guidance in the selection of suitable entrepreneurs and appropriate practical training can develop successful entrepreneurs.

Feature of SIET (Small Industry Extension Training Institute) selection test programmes are psychological tests like TAT, Risk-taking, Personal efficacy;
2) Finding out the socio-economic/educational background of the candidate and 3) personal interviews.

Techniques of identification:

Analysis of application Blank: - Application comprises of the questions related to educational background, family background, previous experience, social participation and level of aspiration of the individual. Applications elicit basic information about the back-ground and experience of the candidate. At successive stages, applications are evaluated on the basis of behavioral science techniques. Testing is meant to measure candidate's motivation to achieve, the capacity to take risks and resolve problems, the extent of their positive self image and their interest in setting up business.

A rigorous screening is being done on the basis of the analysis of the application blank.

TAT (Thematic Appreciation Test): -It is a semi-projective test in which six selected pictures are shown to the individual for a short period of 30 seconds each. He is then asked to write a story about each picture within a time limit of 5 minutes. It is assumed that careful analysis of this test may give a clear indication of his need for achievement, power and motivation.

Ring toss game: - The ring toss game is played with 3 rings and 1 peg. The distance from the peg is marked and the respondents are asked to select any distance for throwing all the 3 rings. The success is that at least 2 rings should be pegged. The risk propensity of the respondents is scored according to the distance chosen and the amount of risk perceived by the respondents from that distance.

Locus of control: - This test gives about the Personal efficacy. The locus of control is measured by administering a questionnaire consisting of 29 pairs of statements representing external and internal locus of control. The respondents are asked to choose one from each pair. They are given scores for their choice. It is assumed that a high score on internal locus of control is indicative of entrepreneurial behavior.

Group Planning Exercise: - Stimulated exercises can be used to observe the group behavior of an individual. A task is given to the candidates and the objectives to be accomplished are explained to them. Resources available are described and a time frame is provided to perform the task. The task in the simulated exercise is so designed that its performance elicits observable behavior of the participants. While the task is performed, two observers score the observed behavior as related to leadership, team spirit, commitment to tasks, planning and

organizing ability and decision-making ability.

Personal Interview: - The personal interview is conducted to assess the knowledge, interest and skill required by the potential entrepreneurs.

There are some drawbacks in this selection process and procedure. In the actual practice has to undergo further refinement, while the principle of entrepreneur selection has found a foot-hold.

Above training programs are conducted by SISI, TCO (Technical Consultancy Organization like APITCO (Andhra Pradesh Industrial Technical Consultancy Organization), KITCO, UPITCO etc., in collaboration with SIET Institute, SBI.

Punjab experience: - The survey revealed that 50% are motivated by their own ideas. 80% were happy with their ventures and ready to convince other persons to become entrepreneurs.

Gujarat experience: - In 1970 Gujarat Entrepreneurship Development Programme was started by the State government. Basic assumption was that the entrepreneurship can be induced or developed. Potential entrepreneurs are identified by industries certain tests designed by Behavioral Sciences Centre, New Delhi. 66% were successful in setting up enterprises.

1.0 Emergence of first generation entrepreneurs:

A first generation entrepreneur is one who starts an individual unit by innovative skills. He is essentially an innovator combining different technologies to produce a marketable product or service.

- Entrepreneurship did not grow early in India due to various reasons like lack of capital, lack of political unity, network of custom barriers, colonial power etc.
- East India Company probably gave new stimulus to Indian business people, especially Parsis, by accelerating the export of raw materials and import of finished goods. The first ship building company was started in Surat where from 1673 onwards Parsis built vessels for East India Company. Lowjee Nushirvan belonging to Wadia family led many leading ship builders for East India Company.
- The managing agents, the businessmen operating in the Agency houses, are the real entrepreneurs of India.
- The first cotton mill was set up by a Parsi- Cowasjee Nanabhoy Davar in Bombay.
- J N Tata gave lead in steel industry in the last quarter of 19th century.
- In Ahmedabad a second largest textile mill was started by Ranchhodlal Chhotalal belonging to a Brahmin family.
- After First World War cement and sugar industries experienced fast progress. A group of self made entrepreneurs began to emerge who by ploughing back their profits into their small workshops, built up larger industrial establishments.

- During Second World War entrepreneurs got many incentives for setting up new industries.
- After independence Government of India devised schemes for balanced and mixed economy. Under the five year plans the government laid emphasis on the growth of small scale industries in cities, towns and villages which led to the rapid emergence of many entrepreneurs.
- Public sector undertakings contributed to the maximum extent to the growth of economy and development of many ancillary industries.
- The advancements in the fields of electronics, communications and computers stimulated many first generation entrepreneurs.

2.1 Entry barriers:

Entry barriers are those forces limiting access to identified business opportunities and capitalization on these opportunities. These are:

1. A cultural bias in identifying and managing the entrepreneurial development process.
2. Insufficient market information and industry data
3. Limited effectiveness of infrastructural base
4. Existence of visible and invisible obstacles to entry of specific social group (ex. Women) into business.
5. Hostile environment
6. Limited access to technology

2.0 Environmental influence:

Modern business is treated as a social and economic institution and is affected by the political, social and economic forces. It is the environment which regulates the entrepreneurial activities. The environment can be classified as

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. Political <ul style="list-style-type: none"> • Political atmosphere • Leadership | <ol style="list-style-type: none"> 2. Economic <ul style="list-style-type: none"> Economic policies Labor Trade Tariffs Incentives Subsidies |
| <ol style="list-style-type: none"> 3. Social <ul style="list-style-type: none"> • Consumer Labor • Attitudes • Opinions • Motives | <ol style="list-style-type: none"> 4. Technological <ul style="list-style-type: none"> Competition & risk Efficiency Productivity & profitability |
| <ol style="list-style-type: none"> 5. Legal | <ol style="list-style-type: none"> 6. Cultural |

Rules & regulations

Structure

Aspirations & values

3.1 Factors influencing entrepreneurship:

1 Socio-demographic variables

- Educational & technical qualifications
- Emigration
- Family background
- Previous occupation

2. Economic variables

Ancestral property
Prior income
Initial investment
Profit utilization
Level of living

3. Variables of system linkages

- Contacts at higher social & govt. levels
- Availability of technical advice
- Mutual help
- Political affiliation
- Social participation
- Personal training

4. Latent characteristics

Leadership qualities
Innovativeness
Risk bearing
Self reliance
Eagerness to evaluate
enterprise & fix long
& short term goals

3.0 Development of women entrepreneurs:

- Women constitute about 50% of the world population.
- Women have been victims of social prejudices and assumptions.
- In traditional societies, women had been confined to the four walls of home, children, household affairs and family rituals and customs.
- In recent years, women have been in the forefront in different walks of life and competing successfully with men despite the social, psychological and economic barriers because of education, political awakening legal safe guards, urbanization, social reforms etc.
- In the 7th five year plan, a special chapter on women's development has been included giving the plan of action for "integration of women in development".
- The new industrial policy of Govt. of India has specially highlighted the need for conducting training programs for women.
- The training programs should be reoriented to include imparting new skills in various areas rather than emphasizing only on female oriented courses like stitching, embroidery, household decoration etc.
- With the spread of education and new awareness, women entrepreneurs are entering the fields of higher levels of 3 Es – Engineering, Electronics & energy. Today no field is unapproachable to women.
- The Govt. of Kerala provides a lot of assistance to women entrepreneurs in Kerala through agencies like KITCO, DICs. The areas of assistance include
 - Preparation of project reports
 - Meeting the cost of machinery and buildings
 - Training & hiring managerial personnel
 - Sales tax exemption for 6 years

Meeting 100% cost of technical expertise

- The factors encouraging women to become entrepreneurs are:
 1. Pull factors
 - Women choose a profession as a challenge and adventure
 - Urge to do something new
 - Liking for business and to have independent occupation
 2. Push factors
 - Takes up enterprise to get on financial difficulties
 - Thrusted responsibility due to family circumstances

4.1 Problems of women entrepreneurs:

1. Being a woman she has to fulfill her responsibilities towards her family.
2. Bear the attitudes of the society towards her and work under the constraints despite the constitutional and legal equality.
3. Lack of proper training for improving or acquiring the necessary skills.
4. Difficulty in rising finances as they are dependent on men
5. Procedures, rules and regulations discourage them.
6. Difficulty in interacting with government departments
7. Difficulties in marketing and exploitation by middlemen both in purchasing raw materials and selling finished goods.
8. Psycho-social barriers
 - Poor self image
 - Inadequate motivation
 - Discriminating treatment
 - Faulty socialism
 - Cultural values
 - Lack of courage and self confidence
 - Lack of freedom

IDBI (Industrial Development Bank of India): - It has completed Industrial potential surveys of 19 states and union territories. The surveys have yielded number of viable projects ideas. It has also set up Inter Institutional Groups (IIGs), at the state levels.

IDBI has constituted a Technical Assistance Fund for financing wide range of developmental activities. The funds could be utilised for financing a) techno-economic surveys, b) Preparation of project profiles, feasibility studies and project reports, c) Promotion of research, d) Providing technical assistance and expertise by sponsoring services of experts and consulting firms, e) training facilities for the personnel of financial institutions in India and abroad and f) entrepreneurial development programmes and entrepreneurial development institutions.

TCO: - Eight Technical Consultancy Organisations (TCO's) at Cochin, Gauhati (North Eastern) Patna, Kanpur, Hyderabad, Bhubaneswar, Jammu and Kashmir and Calcutta. The TCO's are

primarily catering to the needs of new and small entrepreneurs in the areas such as 1) Identification of Projects, 2) Preparation of Project profiles 3) reports, 4) market studies, 5) Technical and managerial assistance.

Apart from the above TCO's are active in identification of new entrepreneurs and have been providing training facilities to them by conducting Entrepreneur Development Programmes (EDP).

ICICI: - Industrial Credit and Investment Corporation of India as an experienced development bank, the ICICI provides promotional assistance to worthwhile projects. Established in 1973 promotional assistance like identification of new projects on the one hand and location of suitable entrepreneurs on the other. Special consideration is also given to new products and processes.

IFCI (Industrial Finance Corporation of India):-

It is the first development bank established on the 1st July, 1948 under a special Statute, with the object of making medium and long term credits more readily available to industrial concerns in India. Its objectives are a) to fill in gaps in the institutional infrastructure for promotion and growth of industries, b) to provide much needed guidance in project identification, formulation, implementation, operation, monitoring etc., to the new, tiny, small scale or medium scale entrepreneurs and to improve the productivity of human and material resources, giving at the same time a better deal to the weaker and under-privileged sections of the society in consonance with the socio-economic objectives laid down by the Government of India.

Also its efforts are towards a) encouraging the adoption of indigenous technology, reviving sick units in the tiny and small scale sectors, c) self-development and self-employment of unemployed young persons, etc.

IFCI set up in 1973, The Management Development Institute (MDI) at Delhi is an autonomous body, helps in developing managerial skills in various areas of functional management in their respective fields of industry. IFCI gives great importance to professionalised management.

Sources of innovation:

1. Unexpected success, Unexpected failures, Unexpected outside events
2. Incongruity (discrepancy between what is and what it ought to be)
3. Process need
4. Industry and market structure
5. Demographics (Changes in population)
6. Changes in perception
7. New knowledge

Principles of innovation:

Principles of innovation require a few Dos and a few Don'ts.

Dos:

1. Purposeful, systematic innovation begins with an analysis of opportunities

2. Innovation is both conceptual and perceptual. The second imperative of innovation is to go out to look, to ask, and to listen.
3. An innovation to be effective has to be simple and has to be focused.
4. Effective innovations start small. They are not grandiose. They try to do one specific thing.
5. A successful innovation aims at **leadership**.

Don'ts:

1. The first is simply not to try to be clever.
2. Do not diversify. Innovations that stray from core are likely to become diffuse. They remain ideas only.
3. Finally do not innovate for the future. Innovate for the present.

Conditions:

1. Innovation is work. Needs knowledge. Requires great ingenuity.
2. To succeed, innovators must build on their strengths.
3. Finally innovation always has to be close to the market, focused on the market, indeed market driven.

Project *ideas* and Technology

PROJECT: - A project can be defined as a scientifically evolved work plan devised to achieve specific objectives within a specified period of time.

A. project can be considered as a proposal involving capital investment for the purpose of developing facilities to provide goods or services.

A project is a specific activity on which money is spent in the expectation of returns.

The project involves allocation and consumption of resources on one hand and generation of resources, goods, or services on the other hand.

The work plan must lay down clearly

1. Manner in which the pre-determined project objectives are sought to be achieved.
2. The resources which will be consumed in the process of achieving the objectives.
3. The time required for achieving the objectives.

Project classification:

Project can be:

1. Production of goods/services.
2. Increasing the capacity of the existing projects (modernisation).
3. Increasing the productivity of the existing means of production.

Projects may be classified to help in graphically expressing and highlighting the essential features of the project. Generally the projects include all activities which are aimed at

1. Production of goods and services
2. Increasing the capacity of the existing projects

3. Increasing the productivity of the existing means of production

The classification may be as follows:

1. Quantifiable and non-quantifiable Projects

A plausible quantitative assessment of benefits can be made in the case of quantifiable projects where as such assessment can not be made in the case of non-quantifiable projects.

a. Ex: Power generation, industrial development, mineral development -

Quantifiable Projects

b. Education, health, defense - Non-quantifiable Projects

2. Sectoral projects

Agricultural & allied sector

Transport & communication

Irrigation and power

Social service

Industry & mining

Miscellaneous

3. Techno-economic projects

1. Factor intensity oriented classification

Capital intensive

Labor intensive

2. Causation oriented

Demand based

Raw material based

3. Magnitude oriented – Depending on total project investment

Large scale

Medium scale

Small scale

United Nations and its specified agencies use the International Standard Industrial Classification of all economic activities (ISIC) in collection and compilation of economic data. Economic activities are classified into ten divisions. These divisions are sub-divided into ninety sub-divisions. This classification covers the entire field of human economic endeavor. The divisions are:

Division 0 Agriculture, Forestry, Hunting and Fishing

Division 1 Mining and quarrying

Division 2

&

Manufacturing

Division 3

Division 4 Construction

Division 5 Electricity, Gas, Water and Sanitary Services

Division 6 Commerce

Division 7 Transport, Storage and Communications

Division 8 Services

Division 9 Activities not adequately described

Conceiving the Idea/Sources of Ideas:-

The entrepreneur has to first search for a sound and workable business idea and must give a practical shape to it. He should be convinced that the idea is sound and likely to give a reasonable return on his investment.

Project ideas originate from the various sources or due to different reasons like

1. The success story of a friend/relative,
2. Experience of others in manufacture/sale of product,
3. Demand for certain products,
4. Chances of producing a substitute of an article imported for which there is good demand,
5. Visits to trade fairs,
6. Study of Project profiles and industrial potential surveys,
7. Meeting government organizations
8. Review of imports and exports
9. Suggestions of financial institutions and development agencies
10. Possibilities of reviving sick units
11. Unfulfilled psychological needs of customers
12. Economic and social trends
13. Study of outlays of government expenditure
14. New technological developments etc., and of course the motivation, background and skill of the entrepreneur and his associates.

The first and foremost problem of an entrepreneur is to find out a suitable business which can give him a reasonable profit. So, entrepreneur has to first search for a sound or workable business idea and gives a practical shape to his idea of the business. He confronts number of problems and his ultimate success will depend upon his ability and foresight to tackle the various problems with which he will be faced from time to time. Therefore problems before starting his new enterprise.

- Marketability of the product
- Its use (industrial use, domestic use, ancillary)
- Its buyers
- Demand and supply over the last few years to estimate its future demand

Demand of the product to be decided after considering of the anticipated changes in technology.

- Levels of incomes of the people
- Throughout the country
- One or two states or a particular region
- Repeat product (like soap* tooth paste etc.)
- Durable article (like watch, refrigerator etc.)
- Low priced to take advantage of large sales/ fix high price so that he can get fair margin even on a smaller volume of sales. In addition dealers' networks or distribution channels to be properly planned.

The entrepreneur should probe the various consequences of giving a practical shape to his ideas with an analytical mind. He should not only draw upon his past experience but also take advice from his reliable business friends. In addition, it may even be advisable to obtain experts advice from professionals, like bank managers, dealers, commercial consultants, advertising agencies and even auditors.

I. Prospective consumers

- II. Items reserved for small scale units
- III. Attending motivation campaigns
- IV. Industrial Potential Surveys
Government organizations like S.I.D.C.,
S.I.S.I., District Industries Centre, S.I.E.
Centers, F.D.C. both at all India and
State levels have identified viable projects
and prepared District Industrial Potential Surveys.

Some abbreviations

D.C. (SSI)	- Development Commissioners - Small Scale Industries
T.C.O.	- Technical Consultancy Organization
I.D.B.I.	- Industrial Development Bank of India (TAF - Technical Assistance Fund)
I.I.G	- Inter Institutional Groups
TCO	- Technical Consultancy Organization in various states (like APITCO - Andhra Pradesh Industrial Technical Consultancy Organization)
I.C.I.C.I.	- Industrial Credit and Investment Corporation of India (Established in 1973)
MDI	- Management Development Institute established in 1973
SIET Institute	- Small Industry Extension Training Institute
SISI	- Small Industry Services Institute
SSIDO	- Small Scale Industrial Development Organization
NPC	- National Productivity Council
NCAER	- National Council of Applied Economic Research
TECSOK	- Technical Consultancy Organization of Karnataka
ISIC	- International Standard Industrial Classification of UNIDO

Difference between a Trade Fair and an Exhibition:-

The word fair is used to describe a general and large international event which admits a wide range of goods, which the exhibition connotes a national event or a specialized fair. Significance of Trade Fair:-

1. Meeting a large number of buyers from different countries at one place, and exchanging views on business environment
2. Assessing the market trend i.e., market in terms of demand potential and type of products required
3. Assessing attitudes of the competitors in a particular product or marketing area
4. Comparing the prices and quality of similar products
5. Establishing personal contacts with the dealers/importers/customers of the product and organizing trade negotiations
6. Assisting the exhibitors to acquire knowledge of the latest scientific techniques of production and marketing, particularly by their competitors
7. Publicity economic achievements of a country and expand export.

Types of Trade Fairs:- These are theme based and purpose based and

categorized on the basis of general and specialized fairs. Some of the trade fairs are HANOVER Engg. Fair, COLOGNE Fair for international clothing machines, MUNICH Fair for cosmetics ANUGA Food Fair at COLOGNE held in West Germany annually. About 2000 trade Fairs and exhibitions are held annually all over the world.

Trade Fairs in India:-

The concept of trade fairs in India may be traced back to the institution of melas in ancient times. A true beginning towards the promotion of a fair-culture in India started with Asia 1972 held at Pragati Maidan (an area of 120 acres of garden land located in the centre of the city) in Delhi. It had impressive halls, restaurants, shopping arcades, auditoriums, museums, art gallery, cinema, administrative offices, including banks, telex and postal facilities, customs clearance arrangements, travel agencies, railway sidings, warehouses and a comprehensive, security and public address systems. India International Trade Fair 1981 and over 40 countries took part.

Trade Fair Authority of India (TFAI) established in 1976 by Government of India not only helps Indian exporters to participate in international trade fairs/exhibitions but organizes wholly Indian exhibitions in several countries with participation by Specialized Commodity Boards, the Trade Development Authority and even some industry and trade associations in collaboration with various agencies. On the recommendations of the Trade Fair Authority of India the RBI releases necessary foreign exchange required by export promoters to visit/participate in international trade fairs.

2.1 Stimulating the flow of ideas:

To help conception of ideas which can be turned into a profitable product, the following steps may be useful.

1. SWOT analysis:

The introspection into the organization's Strengths, Weaknesses, Opportunities and Threats facilitates the generation of new ideas.

2. Clear articulation of objectives:

- Cost reduction
- Productivity improvement
- Increase in capacity utilization
- Improvement in contribution margin

3. Fostering a conducive climate:

- Providing unconstrained environment
- Introduction of suggestion schemes

2.2 Evaluation of ideas:

Evaluation and preliminary screening are needed to eliminate ideas which prima facie are not promising. The following aspects will be helpful in evaluating the ideas in the initial stages.

1. Compatibility with promoter

- Interest
- Personality
- Resources

2. Consistency with government priorities
 - Consistent with national goals and priorities
 - Environmental effects
 - FE requirements
 - licenses
3. Availability of inputs
 - Capital requirements
 - Technical know-how
 - Raw materials
 - Power requirements
4. Adequacy of market
 - Competitors and market share
 - Export market
 - Price profile
 - Sales and distribution system
 - Consumption patterns
5. Reasonableness of cost
 - Cost material inputs
 - Labor costs
 - Economies of scale
 - Overheads
6. Acceptability of risk level
 - Vulnerability to business cycles
 - Technological changes
 - Competition from substitutes, imports
 - Government controls over price and distribution

2.21 Project rating index:

Preliminary evaluation may be streamlined by a rating index which can be worked out as follows.

Steps:

1. Identify factors relevant for project rating
2. Assign weights to these factors reflecting the relative importance
3. Rate the project proposal on various factors using a suitable scale say 5 point scale
4. For each factor multiply the factor rating with factor weight to get the factor score
5. Add all the factor scores to get the over all project rating index.

Construction of rating index

Factors	Factor weight	Rating					Factor score
		VG 5	G 4	A 3	P 2	VP 1	
Input availability	0.25			X			0.75
Technical know-how	0.10		X				0.40
Reasonableness of cost	0.05		X				0.20

Adequacy of market	0.15	X	0.75
Complimentary relationship with other products	0.05	X	0.20
Stability	0.10	X	0.40
Dependence on firm's strength	0.20	X	1.0
Consistency with govt. policies	0.10	X	0.30
Total	1.0		4.00

3.0 Manufacturing Process / Technology:

For manufacturing a product, often two or more alternative technologies are available. For Example:

Steel: Bessemer process or open hearth process

Cement: Dry process or wet process

Hence it is necessary to select an appropriate technology suitable to local economic, social and cultural conditions. Technology selected must also be suitable for the utilization of local raw materials, manpower, and catering to the basic needs.

3.1 Choice of Technology:

Factors influencing the selection of technology:

1. Plant capacity
2. Principal inputs
3. Investment out lay and product cost
4. Use by other units
5. Product mix
6. Latest developments
7. Ease of absorption

3.2 Acquiring Technology:

Technology may either be developed internally or acquired from others who have successfully implemented and been using profitably.

The various methods of acquiring technology are

1. Technology licensing
 - License to use the patented technology on mutually agreed basis
 - Package has to be disaggregated into its components
 - Clear terms and conditions like extent of technology, cost, guarantees provided, duration of license, purchase of intermediate products, components and other inputs, use of trade name etc.
2. Purchase of technology (Transfer of technology)
 - Suitable where technology is stable in the future
 - No further assistance is needed from the seller of technology
 - Capability to absorb technology and maintain study growth.
3. Joint venture arrangement (Collaboration)
 - Technical collaboration
 - Technical and Financial collaboration
 - Buy back arrangements

- Equity holding motivates the supplier in transferring the improvements promptly.

3.3 Collaborative interaction for Technology Development:

- Sharing investment costs
- Exchanging expertise and technological knowledge
- Joint Research and development efforts for mutual benefit
- Expanding market share and competitive advantage

Unit-III

Project formulation, Analysis of market demand, Financial and profitability analysis and Technical analysis. Project financing in India.

Project Formulation

Till recently, in our country much attention was not paid towards preparation of preliminary feasibility and detailed project reports. Most of the important projects were designed with the help of the foreign collaborators in one form or other. If the project work is done intelligently it will throw up technological research problems the solutions to which would promote accelerated development.

In the formulation of any project an important phase is pre-investment phase. The phase consists of the period from the conception of an idea until the final analysis of the necessary elements in order to decide whether the project should be executed or not.

Elements of Project Formulation Techniques: -

Project formulation is by itself an analytical management aid. Project development throws up data in a constant stream. The project formulation team has to evaluate alternative approaches and to arrive at the most effective decision either on its own or with the help of the project sponsoring body.

The aim always is to achieve the project objectives with the minimum expenditure of resources.

Project formulation divides the process of project development into **seven distinct and sequential stages**. The stages are

1. Feasibility analysis
2. Techno-economic analysis
3. Project design and network analysis
4. Input analysis
5. Financial analysis
6. Social cost-benefit analysis and
7. Project appraisal

1. Feasibility analysis:-

The purpose of analysis is to examine the desirability of investing in pre-investment studies. For this purpose examine the project idea in the light of internal and external constraints. Internal constraints are the limitations of the project sponsoring and project implementation body. External constraints are due to the characteristics of the environments.

When a project idea is taken up for development, three situations arise, appear to be a) feasible b) not feasible c) available data may not be adequate for arriving at a reasonable decision which requires additional investment and time.

2. Techno-economic analysis:-

It is concerned with the identification of the project demand potential and the selection of the optional technology which can be used to achieve the project objectives. Project demand is a critical determinant of the optional size of the project. Project size in its own turn determines the technology which will be appropriate to a particular project situation.

Technology includes methodology or the process where technical operations are not involved. Market analysis has to be followed by a detailed search for alternative technologies which can be used to achieve the project objectives.

Techno-economic analysis gives to the project individuality and sets the stage for detailed design development.

3. Project design and Network Analysis: – Project design defines the individual activities and their inter-relationship with each other. This is most inter-relationship with each other. This is most conveniently expressed in the form of a network diagram. This is concerned with the development of the detailed work plan of the project and its time profile.

4. Input Analysis: –It concerns with what the project will consume both during the construction phase as also the normalisation phase.

The objective is to identify and quantify the project inputs and to assess the feasibility of a sustained supply of these inputs all through the effective life span of the project. Inputs are material and human resources.

Input analysis considers the recurring as well as non-recurring resources requirements of the project and evaluates the feasibility of the project from the point of view of the availability of these resources.

5. Financial analysis:–It concerns itself with the estimation of the project costs, project operating costs and project funds requirements. It also involves the appraisal of financial characteristics of the project, so as to establish the merits and demerits of the project as compared to other investment opportunities. A large number of financial analytical aids developed are: present worth technique, the cost volume–profit analysis and ratio analysis. The uncertainties have to be taken into account.

6. Cost Benefit analysis:–The cost–benefit analysis takes into account not only the direct costs and benefits which will accrue to the project implementing body but also the total costs which all entities concerned with the project will have to bear and the benefits which will be enjoyed by all such entities.

Idea is to evaluate the project in terms of absolute costs and benefits rather than in terms of apparent costs and benefits.

7. Pre–investment Appraisal: – It is the process of consolidating the above i.e., feasibility analysis, techno–economic analysis, Project design and network analysis, input analysis, financial analysis and social cost–benefit analysis so as to give the investment proposition a final and formed shape.

The sum total of the pre-investment appraisal is to present the project idea in a form in which the project sponsoring body the project implementing body and the outside agencies can take an investment decision regarding the proposals.

1.0 Analysis of market Demand: Market and demand analysis is concerned with two broad issues

1. What is likely aggregate demand for the product / service?
2. What share of the market will the proposed project enjoy?

Factors to be considered for getting the answers for the above:

1. Patterns of consumption growth
2. Income and price elasticity of demand
3. Composition of the market
4. Nature of competition
5. Availability of substitutes
6. Distribution channels

3.1 Steps in market analysis:

1. Situation analysis and specification of objectives:
 - ❖ Talk to customers, competitors, middlemen and others
 - ❖ For carrying out market survey spell out the objectives clearly
 - ❖ Questionnaire can help gathering the information in a way relevant for forecasting the demand
2. Collection of secondary inputs:
 - ❖ Secondary information is the one which was gathered in some other context and is available readily for the present consideration
 - ❖ Primary information is the one, which is collected for the first time to meet the specific purpose on hand.

❖ Secondary information forms the basis and starting point for the market and demand analysis.

❖ General sources of secondary information:

Censuses of India, National sample survey reports, Plan reports, India year book, Statistical year book, Economic survey, Guide lines to industries, Annual survey of industries, Publications of advertising agencies, Monthly bulletin of RBI, etc.

Annual reports of association of Indian automobile manufacturers Journals of industry associations.

❖ The relevance, reliability, accuracy are to be carefully studied in the information available in the secondary information.

3. Conduct of market survey:

❖ Secondary information may not provide a comprehensive basis often thus necessitating gathering primary information through market survey.

❖ Census survey: Entire population is covered. It is suitable for intermediate goods, investment goods - where the number is less.

❖ Sample survey: A sample of the population is contacted or observed. Inferences are made on the basis of the information gathered from the sample. Ex:

- Total demand & rate of growth of the demand,
- Demand in different segments of the market,
- Income & price elasticity of the demand,
- Motives for buying, purchasing plan and intentions,
- Satisfaction with existing goods,
- Unsatisfied needs,
- Attitudes towards various products,
- Distribution trade practices and preferences,
- Socio-economic characteristics of buyers

1.1.1 Steps in conducting Sample survey:

- i. Define the target population
- ii. Select the sampling scheme and sample size
- iii. Develop the questionnaire
- iv. Recruit and train the field investigators
- v. Obtain the information as per questionnaire from the sample respondents
- vi. Scrutinize the information gathered
- vii. Analyze and interpret the information

1.1.2 The results of the information can be vitiated by

1. Non representativeness of the sample
2. Questions lack precision and accuracy
3. Failure of respondents to comprehend the questions
4. Deliberate distortions in the answers given by the respondents
5. Inept handling of the interviews
6. Cheating on the part of the investigators
7. Incorrect and inappropriate analysis and interpretation of data

1.1.3 Problems:

1. Heterogeneity of the country
2. Multiplicity of the languages
3. Design of questionnaire

4. Characterization of the market:

❖ The market may be described on the basis of the information gathered in the survey

1. Effective demand in the past and present:

Apparent consumption = Production + Imports – Exports – Change in stock level

Consumption of the product by producers and effect of abnormal factors are to be adjusted.

In a competitive market effective demand and apparent consumption are equal.

2. Breakdown of the demand:

Aggregate demand may be broken down into demand for different segments of the market

Market segments may be defined by

1. Nature of the product
2. Consumer group
3. Geographical division

Segmental information is helpful because the nature of the demand tends to vary from one segment to the other

3. Price:

1. Manufacturer's price quoted FOB(Free On Board)
2. CIF price (Cost, Insurance, Freight)
3. Landed price for imported goods
4. Average whole sale price

4. Methods of distribution and sales promotion

Distribution varies in the nature of the product

Different distribution channels may be used for a given product

Advertisement, discounts, Gifts scheme vary from product to product

5. Consumers:

Demographic & Sociological	Attitudinal
Age Profession	Preferences
Sex Residence	Intentions
IncomeSocial background	Habits
	Attitudes
	Responses

6. Supplies and Competition:

Existing sources - Indigenous / imported

Location, present production Capacity

Planned expansion, capacity utilization level

Bottlenecks in production, Cost structures,

Quantity, Quality, Promotional efforts

7. Government policy:

Production targets in National plan

Import & export trade constraints

Import duties and incentives

Excise duties and sales tax , industrial licensing

Credit controls, preferential purchasing

Financial regulations, subsidies , Penalties

5. Demand Forecasting:

• Quantitative methods:

1. Jury of executives opinion method

2. Delphi method: involves converting the views of group experts , who do not interact face to face , into a forecast through an iterative process.

• Time series method:

1. Trend projection method
2. Exponential smoothening method
3. Moving averages method

- Casual models:

These are based on the cause-effect relationship.

1. Chain ratio method – Applies a series of factors for developing a forecast.
2. Consumption method
 - Income elasticity of demand
 - Price elasticity of demand
3. End use method – suitable for intermediate products
4. Leading indicator method – Observed changes in leading indicators are used to predict the changes in lagging variables.
5. Econometric method – Estimating quantitative relationship derived from economic theory.
6. Market planning:
 1. Pricing
 2. Distribution
 3. Promotion
 4. Service

FINANCIAL PROFITABILITY:- The estimates of financial profitability may be prepared along the following lines:

- A. Cost of production
- B. Total administrative expenses
- C. Total sales expenses
- D. Royalty and know-how payable
- E. To cost of production (A+B+C+D)
- F. Expected sales
- G. Gross profit before interest
- H. Total financial expenses
- I. Depreciation
- J. Operating profit (G–H–I)
- K. Other income
- L. Preliminary expenses written off
- M. Profit/loss before taxation (J+K–L)
- N. Provision for taxation
- O. Profit after tax (M–N)

Less Dividend on

- Preference capital
- Equity capital

P. Retained profit

Q. Net cash accrual (P+I+L)

A. Cost of Production: – This represents the cost of materials, labour, utilities and factory overheads as calculated earlier.

B. Total administrative expenses: – This consists of

(i) Administrative salaries, (ii) remuneration to directors, (iii) Professional fees, (iv) light, postage, telegrams, and telephones, and office supplies (stationery, printing, etc.) (v) Insurance and taxes on office property, and (vi) miscellaneous items.

C. Total sales expenses:– The expenses included under this head are:

i) commission payable to dealers, (ii) packing and forwarding charges, (iii) salary of sales staff (which may be increased at 5 percent per annum), (iv) sales promotion and advertising expenses, and (v) other miscellaneous expenses.

The selling expenses depend mainly on the nature of industry and the kind of competitive conditions that prevail. Typically, selling expenses vary between 5 and 10 percent of sales. The experience of similar firms in the industry may be used as basic guideline.

D. Royalty and know-how payable: – Royalty and know-how payable annually may be shown here. The royalty rate is usually 2-5 percent of sales. Further, royalty is payable often for a limited number of years, say 5 to 10 years.

E. Total cost of Production: – This is simply the sum of cost of production, total administrative expenses, total sales expenses, and royalty and know-how payable.

F. Expected sales: - The figures of expected sales are drawn from the estimates of sales and production prepared earlier in the financial analysis and projection exercise.

Typically, the starting point for profitability is the forecast for sales revenues. In estimating sales revenues, the following considerations should be borne in mind:

1. It is not advisable to assume a high capacity utilization level in the first year of operation. Even if the technology is simple and the company may not find technical problems in achieving a high rate of capacity utilization in the first year itself, there are likely to be other constraints like raw material shortage, limited power, marketing problems, etc. It is sensible to assume that capacity utilization would be somewhat low in the first year and rise thereafter gradually to reach the maximum level in the third or fourth year of operation. A reasonable assumption with respect to capacity utilization is as follows: -

40 - 50 percent of the installed capacity in the first year,

50 - 80 percent in the second years and 80 - 90 percent from the third year onwards.

2. It is not necessary to make adjustments for stocks of finished goods. For practical purposes it may be assumed that production would be equal to sales.

3. The selling price considered should be the price realizable by the company net of excise duty. It shall, however, include dealers' commission which is shown as an item of expense (as part of sales expenses).

4. The selling price used may be the present selling price--it is generally assumed that changes in selling price will be matched by proportional changes in cost of production (This assumption is commonly used by financial institutions. However, it is not very convincing). If a portion of production is saleable at controlled price, take the controlled price for that portion.

Sales and production are closely inter-related. Hence they may be estimated together. For this purpose the format XII A of the application form need by all-India financial institutions in India, may be employed.

(Details may be furnished separately for each product and until the plant reaches maximum capacity utilization)

		Product 1				Product 2			
		1 yr.	2 yr.	3 yr.	4 yr.	1 yr.	2 yr.	3 yr.	4 yr.
1	Installed Capacity (qty/day/annum)								
2	No. of working days								
3	No. of shifts								
4	Estimated production per day (qty.)								

5	Estimated annual production (qty.)								
6	Estimated output as % of plant capacity								
7	Sales (qty.) (after adjusting stocks)								
8	Value of sales (in '000 of Rs.) Product								
	i)								
	ii)								
	iii)								

Note: - Production in the initial period should be assumed at a reasonable level of utilization of capacity increasing gradually to attain full capacity in subsequent years.

G. Gross profit before interest: This represents the difference between expected sales and total cost of production.

H. Total financial expenses: - Financial expenses consist of interest on term loans, interest on bank borrowings, commitment charges on term loans, and commission for bank guarantees. The principal financial expenses, of course, are interest on term loan interest on bank borrowings.

In estimating the interest on term loans, two points should be borne in mind: i) Interest on term loans is based on the present rate of interest charged by the term loan lending financial institutions and commercial banks, ii) Interest amount would decrease according to the repayment schedule of term loan.

The interest on bank borrowings may be estimated as follows:

- (i) determine the total requirement of the working capital,
- (ii) find out the quantum of bank borrowing that would be available against the total working capital requirement, and
- { iii) calculate to the interest charges on the basis of the prevailing interest rates.

I. Depreciation: - This is an important item, particularly for capital-intensive projects. In figuring out the depreciation charge, the following points should be borne in mind:

1. Contingency margin and pre-operative expenses provided in estimating the cost of project should be added to fixed assets proportionately to ascertain the value of fixed assets for determining the depreciation charge.
2. Preliminary expenses in excess of 2.5% of the project cost (excluding working capital margin) should be added to fixed assets proportionately to ascertain the value of fixed assets for determining the depreciation charge.
3. The Income Tax Act specifies that the written down value method should be used for tax purposes. It further specifies the rate of depreciation applicable to different kinds of assets.
4. For company law (financial reporting) purposes, the method of depreciation may be either the written down value (WDV) method or the straight line (SL) method. From 1988 onwards the depreciation rates under the Companies Act have been delinked from those under the Income Tax Act. The key depreciation rates under the Companies Act are as follows:

		In percentage terms					
		Single Shift		Double Shift		Triple Shift	
		WDV	SL	WDV	SL	WDV	SL
1	Buildings (other than factory buildings)	5.00	1.63				
2	Factory buildings	10.00	3.34				
3	Plant and machinery (general rate)	15.00	5.15	22.50	8.09	30.00	11.31

According to the written down value method, the depreciation charge is a certain percentage of the written down value of the asset.

In general terms, the depreciation charge for the n^{th}

n^{th} year is:

$$D_n = I(1-d)^{n-1}d$$

Where D_n = depreciation charge for the n^{th} year

I = initial cost

d = depreciation rate

K. Other Income: - Other income, if any, details have to be given. This represents income arising from transactions not part of the normal operations of the firm. Examples of such transactions are: sale of machinery, disposal of scrap, etc. Except disposal of scrap, which can be reasonably anticipated and estimated, the effects of other non-operating transactions can hardly be estimated. Of course, when non-operating transactions result in a deficit, other income would be negative—put differently, there will be a non-operating loss.

L. Write-off of Preliminary Expenses: - Preliminary expenses up to 2.5% of the cost of project or capital employed whichever is higher, can be amortized in ten equal annual installments.

M. Profit/Loss before Taxation: - This is equal to: operating profit + other income - write-off of preliminary expenses.

N. Provision for Taxation: - To figure out the tax burden, sound understanding of the Income Tax Act -- a complicated legislation -- and relevant case laws is required. While calculating the taxable income, a variety of incentives and concessions have to be taken into account. Once the taxable income, as per the Income Tax Act, is calculated, the tax burden can be figured out fairly easily by applying the appropriate tax rates.

O. Profit after Taxation: - This is simply profit/loss before taxation minus provision for taxation. A part of profit after tax is usually paid out as dividend--dividend on preference capital and dividend on equity capital.

P. Retained Profit: - The difference between profit after tax and dividend payment is referred to as retained profit. It is also called ploughed back earnings.

Also Depreciation and Preliminary Expenses Written Off are to be added to Retained Profit.

Q. Net Cash Accrual: - The net cash accrual from operations is equal to: retained profit + depreciation + write-off of preliminary expenses + other non-cash charges.

R. Total Financial Expenses: - Financial Expenses includes Interest on term loans, Interest on borrowings for working capital and Guarantee commission.

Form XII of the application form used by all-India financial institutions for preparing the estimates of working results or profitability projections and the statement should be prepared for ten years giving detailed working shall be provided for calculation of depreciation (straight line and income tax method), interest, taxation, etc.

COST OF PROJECT: - Conceptually the cost of project represents the total of all items of outlay associated with a project which are supported by long-term funds. It is the sum of the outlays on the following: -

1. Land and site development
2. Buildings and civil works
3. Plant and machinery
4. Technical know-how and engineering fees
5. Expenses on foreign technicians and training of Indian technicians abroad
6. Miscellaneous fixed assets
7. Preliminary and capital issue expenses

8. Pre-operative expenses
9. Provision for contingencies
10. Margin money for working capital
11. Initial cash losses

1. Land and site development:- It includes i) Basic cost of land ii) Premium payable on leasehold iii) Cost of leveling and development iv) Cost of laying approach roads and internal roads v) Cost of gates and vi) Cost of tube wells
The cost of land varies considerably from one location to another.

2. Buildings and Civil Works: - It covers the followings

— Buildings for the main plant and equipments

— Buildings for auxiliary services like steam supply, workshops, laboratory, water supply etc.

— Godowns, warehouses, and open yard facilities – Non-factory buildings like canteen, guest houses, time office, and excise house etc.

— quarters for essential staff

— Silos (or bins for raw material storage), tanks, wells,

basins, cisterns, hoppers, bins and other structures

necessary for installation of plant and equipment Garages

- Sewers, drainage, etc.

The cost of buildings and civil works depends on the kinds of structures required which, in turn, are dictated largely by the requirements of the manufacturing process.

3. Plant Machinery: - It consists of

-- Cost of imported machinery: - This is the sum of

i) FOB (free on board) value, ii) shipping, freight, and insurance cost, iii) import duty, and iv) clearing, loading, unloading and transportation charges.

— Cost of indigenous machinery: - This consists of i) FOR (free on rail) cost, ii) sales tax, octroi, and other taxes, if any, and iii) railway freight and transport charges to site.

-- Cost of stores and spares

— Foundation and installation charges

The cost of plant and machinery is based on the latest available quotation adjusted for possible escalation.

4. Technical Know—how and Engineering Fees:- Often it is necessary to engage technical consultants or collaborators from India and/or abroad for advice and help in various technical matters like preparation of project report, choice of technology, selection of plant and machinery, detailed engineering, and so on. While the amount payable for obtaining technical know—how and engineering services for setting up the project is a component of project cost, the royalty payable annually, which is typically a percentage of sales, is an operating expense taken into account.

5. Expenses on Foreign Technicians and Training of Indian Technicians Abroad:-

For Foreign Technicians, expenses on their travel, boarding, and lodging along with their salaries and allowances must be shown here. Likewise, expenses on Indian technicians who require training abroad must also be included here.

6. Miscellaneous Fixed Assets: - Fixed assets and machinery which are not part of the direct manufacturing process may be referred to as miscellaneous fixed assets. They include items like furniture, office machinery and equipment, vehicles, railway siding, diesel generating sets, transformers, boilers, piping systems, laboratory equipments, workshop equipments, effluent treatment plant, fire fighting equipments, and so on. Expenses incurred for procurement or use of patents, licenses, trade marks, copyrights, etc, and deposits made with the electricity board may also be included here.

7. Preliminary and Capital issue Expenses: - Expenses incurred for identifying the project, conducting the market survey, preparing the feasibility report, drafting the memorandum and articles of association, and incorporating the company are referred to as preliminary expenses. Expenses borne in connection with the raising of capital from the public are referred to as capital issue expenses. These are: underwriting commission, brokerage, fees to managers and registrars, printing and postage expenses, advertising and publicity expenses, listing fees and stamp duty.

8. Pre—operative Expenses: - Expenses of the following types incurred till the commencement of commercial production are referred to as pre—operative expenses.

Pre—operative expenses are directly related to the project implementation schedule. So, delays in project implementation, which are fairly common, tend to push up these expenses. Appreciative of this, financial institutions allow for some delays (20 to 25%) project implementation schedule and accordingly permit a cushion in the estimate for pre—operative expenses.

9. Provision for Contingencies: — Provision for contingencies is made to provide for certain unforeseen expenses and price increases over and above the normal inflation rate which is already incorporated in cost estimates.

10. Margin Money for Working Capital: - A certain part of working capital requirement has to come from long—term sources of finance. The principal support for working capital is provided by commercial banks and trade creditors.

The first part of the above is called Margin money for working capital.

The margin money for working capital is sometimes utilized for meeting over—runs in capital cost. This leads to working capital problem (and sometimes a crisis) when the project is commissioned. To mitigate this problem, financial institutions stipulate that a portion of the loan amount, equal to the margin money for working capital, be blocked initially so that it may be released when the project is completed.

11. Initial Cash Losses: - Most of the projects incur cash losses in the initial years. Yet, promoters typically do not disclose the initial cash losses because they want the project to appear attractive to the financial institutions and the investing public. Failure to make provision for such cash losses in the project cost generally affects the liquidity position and impairs the operations. Hence prudence calls for making a provision, overt or covert, for the estimated initial cash losses.

MEANS OF FINANCE: — To meet the cost of project the following means of finance are available:

1. Share capital
2. Term loans
3. Debenture capital
4. Deferred credit
5. Incentive sources
6. Miscellaneous sources

1. Share Capital:— There are two types of share capital—equity

capital and preference capital. 'Equity capital' represents the contribution made by the owners of the business, the equity shareholders, who enjoy the rewards and bear the risks of ownership. Equity capital being risk capital carries no fixed rate of dividend. 'Preference capital' represents the contribution made by preference shareholders and the dividend paid on it is generally fixed.

2. Term Loans:— Provided by financial institutions and commercial banks, 'term loans' represent secured borrowings which are a very important source (and often the major source) for financing new projects as well as expansion, modernization, and renovation schemes of existing firms. There are two broad types of term loans available in India: 'rupee term loan' and 'foreign currency term loan'. While the former are given for financing land, building, civil works and indigenous plant and machinery and so on, the latter are provided for meeting the foreign currency expenditure towards import of equipments and technical know—how.

3. Debenture Capital:— Akin to promissory notes, debentures are instruments for raising debt capital. There are two broad types of debentures, non—convertible debentures and convertible debentures. Non—convertible debentures are straight debt instruments. Typically they carry a fixed rate of interest and have a maturity period of 5 to 9 years. Convertible debentures, as the name implies are debentures which are convertible, wholly or partly into equity shares. The conversion period and price are announced in advance.

4. Deferred Credit: — Many a time the suppliers of plant and machinery offer a deferred credit facility under which payment for the purchase of plant and machinery can be made over a period of time.

5. Incentive Sources:— The government and its agencies may provide financial support as incentive to certain types of promoters or for setting up industrial units in certain locations. These incentives may take the form of 'seed' capital assistance (provided at a nominal rate of interest to enable the promoter to meet his contribution to the project), or 'capital subsidy' (to attract industries to certain locations), or 'tax deferment or exemption' (particularly from sales tax) for a certain period.

6. Miscellaneous sources: – A small portion of project finance may come from miscellaneous sources like unsecured loans as public deposits, leasing and hire purchase finance. ‘Unsecured loans’ are typically provided by the promoters to bridge the gap between the promoters’ contribution (as required by the financial institutions) and the equity capital the promoters can subscribe to. ‘Public deposits’ represent unsecured borrowings from the public at large. Leasing and hire purchase finance represent a form of borrowing different from the conventional term loans and debenture capital.

Sources of finance for raising capital are:-

1. Equity Capital
2. Preference capital
3. Non-convertible debenture
4. Convertible debentures
5. Rupee term loans
6. Foreign currency term loans
7. Euro issues
8. Differed Credit
9. Bill rediscounting Scheme
- 10 Suppliers line of credit
11. Seed capital assistance
12. Government subsidies
13. Sales tax deferment and exemption
14. Unsecured loans and deposits
15. Lease and hire purchase finance.

1. Equity Capital: - This is the contribution made by the owners of business, the equity shareholders, who enjoy the rewards and bear the risks of ownership. Equity capital offers two important advantages:

- i) It represents permanent capital. Hence there is no liability for repayment,
- ii) It does not involve any fixed obligation for payment of dividends.

The disadvantages of raising funds by way of equity capital are:

- i) The cost of equity capital is high because equity dividends are not tax-deductible expenses,
- ii) The cost of issuing equity capital is high.

2. Preference Capital: - This is like debt capital since rate of preference dividend is fixed. It is similar to equity capital because preference dividend, like equity dividend, is not a tax-deductible payment. Typically, when preference dividend is Skipped it is payable in future because of the cumulative feature associated with it. The, near-fixity of preference dividend payment renders preference capital somewhat unattractive in general as a source of finance. It is, however, attractive when the promoters do not want a reduction in their share of equity and yet there is need for widening the net worth base (net worth consists of equity and preference capital) to satisfy the requirements of financial institutions. In addition to the conventional preference shares, a company may issue Cumulative Convertible Preference Shares (CCPS). These shares carry a dividend rate of 10% (which if unpaid, cumulates) and are compulsorily convertible into equity shares between three and five years from the date of issue.

3 & 4. Debenture Capital: - It is similar to promissory note. In the last few years, debenture capital has emerged as an important source for project financing. There are three types debentures that are commonly used in India:

a) Non-Convertible Debentures (NCDs), Partially Convertible Debentures (PCDs) and Fully Convertible Debentures (FCDs).
NCDs are used by companies for raising debt that is generally retired over a period of 5 to 10 years. They are secured by a charge on the assets of the issuing company.

PCDs are partly convertible into equity shares as per pre-determined terms of conversion. The unconverted portion of PCDs remains like NCD. FCDs, as the name implies, are converted wholly into equity shares as per pre-determined terms of conversion. Hence FCDs may be regarded as delayed equity instrument.

5. Rupee Term Loans: - Provided by financial institutions and commercial banks, rupee term loans which represent secured borrowings are a very important source for financing new projects as well as expansion, modernization, and renovation schemes of existing units. These loans are generally repayable over a period of 8 - 10 years which includes a moratorium period of 1 - 3 years.

6. Foreign Currency Terms Loans: - Financial institutions provide foreign currency term loans for meeting the foreign currency expenditures towards import of plant, machinery, and equipment and also towards payment of foreign technical know-how fees. Under the general Scheme, the periodical liability towards interest and principal remains in the currency/currencies of the loan/s and is translated into rupees at the then prevailing rate of exchange for making

payments to the financial institution.

Apart from approaching financial institutions (which typically serve as intermediaries between foreign agencies and Indian borrowers), companies can directly obtain foreign currency loans from international lenders. More and more companies appear to be doing so presently.

7. Euro issues: - From middle of 1992, a number of companies have been making euro issues. They have employed two types of securities:

Global Depository Receipts (GDRs) and Euro convertible Bonds (ECDs).

Denominated in US dollars, a GDR is a negotiable certificate that represents the publicly traded in local currency (Indian Rupee) equity shares of a non-US (Indian) company (of course, in theory, a GDR may represent a debt security, in practice it rarely does so) GDRs are issued by the Depository Bank (such as the Bank of New York) against the local currency shares (such as Rupee shares) which are delivered to the depository's local custodial banks. GDRs trade freely in the overseas markets.

A Euro convertible Bond (ECB) is an equity-linked debt security. The holder of an ECB has the option to convert it into equity shares at a pre-determined conversion ratio during a specified period. ECBs are regarded as advantageous by the issuing company because i) they carry a lower rate of interest compared to a straight debt security, ii) they do not lead to dilution of earnings per share in the near future, and iii) they carry very few restrictive covenants.

8. Deferred Credit: - Many a time the suppliers of machinery provide deferred credit facility under which payment for the purchase of machinery is made over a period of time. The interest rate on deferred credit and the period of payment vary rather widely. Normally, the supplier of machinery when he offers deferred credit facility insists that the bank guarantee should be furnished by the buyer.

9. Bills Rediscounting Scheme: - Operated by the IDBI, the bills rediscounting scheme is meant to promote the sale of indigenous machinery on deferred payment basis. Under this scheme, the seller realizes the sale proceeds by discounting the bills or promissory notes accepted by the buyer with a commercial bank which in turn rediscounts them with the IDBI. This scheme is meant primarily for balancing equipments and equipments and machinery required for expansion, modernization, and replacement schemes.

10. Suppliers' line of Credit: - Administered by the ICICI, the

Suppliers' Line of Credit is somewhat similar to the IDBI's Bill

Rediscounting Scheme. Under this arrangement ICICI directly pays to the machinery manufacturer against issuance of bills duly accepted or guaranteed by the bank of the purchaser.

11. Seed Capital Assistance: - Financial institutions, through what may be labeled broadly as the 'Seed Capital Assistance scheme', seek to supplement the resources of the promoters of the small and medium scale industrial units which are eligible for assistance from all-India financial institutions and/or state-level financial institutions. Broadly three schemes have been formulated:

i) Special Seed Capital Assistance Scheme: The quantum of assistance under this scheme is Rs.0.2 million or 20% of the project cost, whichever is lower. This scheme is administered by the State Financial Corporations.

ii) Seed Capital Assistance Scheme: The assistance under this scheme is applicable to projects costing not more than Rs.20 million. The assistance per project is restricted to Rs.1.5 million. The assistance is provided by IDBI through state level financial institutions. In special cases, the IDBI may provide the assistance directly.

iii) Risk Capital Foundation Scheme: Under this scheme, the Risk Capital Foundation an autonomous foundation set up and funded by the IFCI, offers assistance to promoters of projects costing between Rs.20million and Rs.150million. The ceiling on the assistance provided between Rs.1.5million and Rs.4million depending on the number of applicant promoters.

12. Government Subsidies: - Previously the central government as well as the state governments provided subsidies to industrial units located in backward areas. The central subsidy has been discontinued but the state subsidies continue. The state subsidies vary between,

5% to 25% of the fixed capital investment in the project, subject to a ceiling varying between RS.0.5million and Rs.2.5million depending on the location.

13. Sales, Tax Deferrals and Exemptions: - To attract industries, the states provide incentives, in the form of sales tax deferrals and sales tax exemptions.

Under the sales tax deferral scheme, the payment of sales tax on the sale of finished goods may be deferred for a period ranging between 5 to 12 years. Essentially it implies that the project gets an interest free loan, represented by the quantum of sales tax deferred, during the deferral period.

Under the sales tax exemption scheme, some states exempt the

payment of sales tax applicable on purchases of raw material, consumables, packing, and processing materials from within the state which are used for manufacturing purposes. The period of exemption ranges from three to nine years depending upon the state and the specific location of the project within the state.

14. Unsecured Loans and Deposits: - Unsecured loans are typically provided by the promoters to fill the gap between the promoter's contribution required by financial institutions and the equity capital subscribed to by the promoters. These loans are subsidiary to the institutional loans. The rate of interest chargeable on these loans is less than the rate of interest on the institutional loans. Finally these loans cannot be taken back without the prior approval of financial institutions.

Deposits from public, referred to as public deposits, represent unsecured borrowing of 2 to 3 years duration. Many existing companies prefer to raise public deposits instead of term loans from financial institutions because restrictive covenants do not accompany public deposits. However, it may not be possible for a new company to raise public deposits. Further, it may be difficult for it to repay public deposits within 3 years.

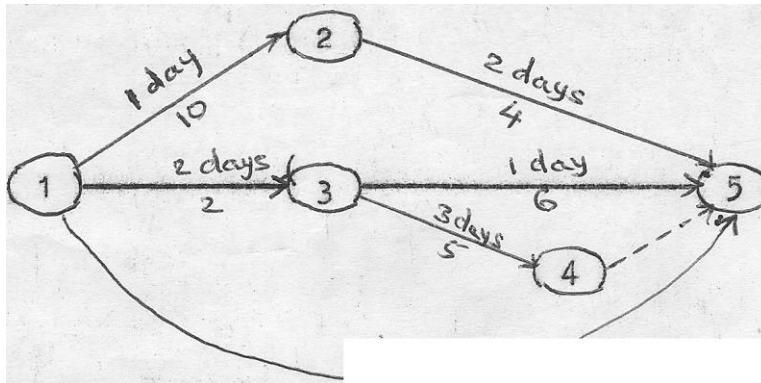
15. Leasing and Hire Purchase Finance: - With the emergence of scores of finance companies engaged in the business of leasing and hire purchase finance, it may be possible to get a portion, albeit a small portion, of the assets financed under a lease or a hire purchase arrangement.

Typically, a project is financed partly by financial institutions and partly through the resources raised from the capital market. Hence, in finalizing the financing schemes for a project, you should bear in mind the norms and policies of financial institutions and the guidelines of Securities Exchange Board of India and the requirements of the Securities Contracts Regulation Act (SCRA).

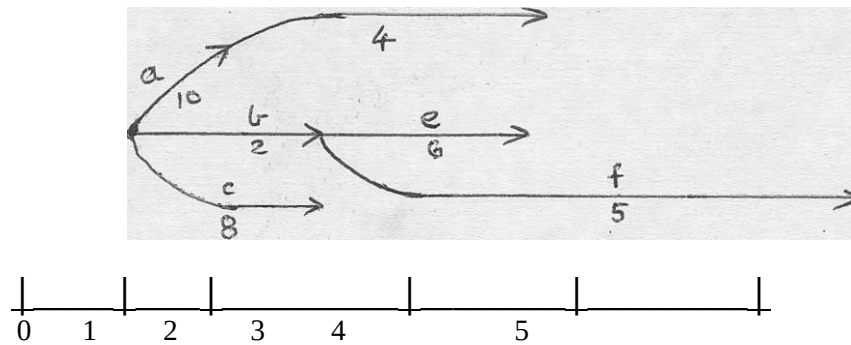
Meeting manpower requirement:-

The manpower required for the activities is indicated by the length of the bar in the bar chart (most suitable for small projects). These bar chart can be used for manpower planning. For large and complex projects Network techniques can be used.

The manpower requirement is shown below the activity arrow. Let us consider a small project for which the network diagram in shown in below:-



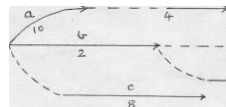
Looking at the manpower requirement for the early start schedule we find that it is 20 for the first day, 14 for the second day, 15 for the third day, 5 for the fourth day, and 5 for the fifth day. Real life situations, however, there may be restrictions on the availability of resources. For example, manpower supply may be limited. When restrictions exist various schedules may have to be considered to find out which one is most appropriate in the light of these restrictions. In the above network diagram, only 12 men are available for the project (manpower resource constraint). The early start schedule of this project is shown as a graph on the horizontal timescale.



Looking at the manpower requirement for the early start schedule we find that it is 20 for the first day, 14 for the second day, 15 for the third day, 5 for the fourth day, and 5 for the fifth day. Obviously, this schedule is unacceptable in view of the manpower constraint.

So, we explore the possibility of shifting activities. Our efforts at shifting activities, keeping the project duration at five days, soon reveals that no schedule is feasible with only 12 men. So we extend the duration of the project, by one day and try various schedules to see whether we can find a feasible schedule. A little juggling of activities shows that a schedule like the one shown as below is feasible – this is the best we can do.

rA



$$\frac{e}{G}$$

5

2—

4

.57

BREAK EVEN POINT: – Break even point which refers to the level of operation at which the project makes neither profit nor incurs loss is calculated.

Dealing with Government and Financial Institutions:–

Before a project actually reaches the resource allocation level it has to pass through several levels of examination, scrutiny, analysis and appraisal. Delays generally take place when a project has to be screened through so many levels, but the worst delays take place on account of the fact that no systematic or scientifically evolved procedure is used either for the preparation or evaluation or appraisal of the project.

The Government official who deals with the project (like State Electricity Board officials, Pollution Control Board officials for issuing No objection Certificate, or obtaining Land use certificate and clearances under Factory Act for building plans/machinery installation clearances from Director, Town and Country Planning/Urban Development Authority, Health Department official etc) who deals with the project, and who has a positive role to play in the final clearance of a project has to be treated as an integral part of the project formulation team (which consists of a team of experts). He has to be exposed to the details of the

project to the same extent as any other member of the team which formulates the pre-investment report.

Financial institutions have to perform dual role. They have to provide institutional support to development activities and secondly they have to identify bankable investment propositions.

No development financial institution advances funds today merely on the basis of credit worthiness of the entrepreneur. Every institution makes its own assessment and appraisal of the investment opportunity and only after satisfying itself about the capacity of the project to repay the investment and also the desirability of what a project will contribute to the overall development of the country, does it part with its funds.

Project formulation team, in effect, tries to bring the project sponsoring authority and the development finance institutions on one wavelength where the parameters of project appraisal laid down by the development finance institutions are integrated into the scheme of development of project ideas and data and other information are collected, evaluated and presented in a form which enables the appraisal team to quickly and efficiently deal with the project.

Some of the financial institutions are ICICI (Industrial Credit & Investment Corporation of India), IDBI (Industrial Development Bank of India), SIDC (State Industrial Development Corporation like APSFC) & SFC (State Financial Corporation like APSFC).

Govt. (Central/State) interventions in all fields of social and economic life based on available knowledge of the conditions and their interrelationship and aimed at the acceleration of development.

National plans are presented with considerable attention and care which identify the priority sectors and production targets for the Country as a whole, and defines the resources mobilization effort which the nation will have to undertake and specify the broad sectoral allocation of resources. And it passes labour laws, controls the prices, exchange regulations etc.

FEASIBILITY STUDIES: – Feasibility report lies in between the project formulation stage and the appraisal and sanction stage. It consists of

1. General information
2. Preliminary analysis of alternatives
3. Project description
4. Marketing Plan
5. Capital Requirements and Costs
6. Operating Requirements and costs and
7. Financial analysis.

1. General Information: - The feasibility report should include an analysis of the industry to which the project belongs. It should deal with the past performance of the industry. Description of the type of industry should also be given.

2. Preliminary analysis of alternatives:- This should contain present data on the gap between demand and supply for the outputs which are to be produced, data on the capacity that would be available from projects that are in production or under implementation at the time the report is prepared, a complete list of all existing plants in the industry, giving their capacity and level of production actually attained, a list of all projects for which letters of intent/licenses have been issued and a list of proposed projects. All options that are technically feasible should be considered at this preliminary stage. The location of the project and its implications should also be looked into. An account of the foreign exchange requirement should be taken. The profitability of different options should also be given. The rate of return on investment should be calculated and presented in the report. Alternative cost calculations
Vis-à-vis return should be presented.

3. Project Description: - The feasibility report should provide a brief description of the technology/process chosen for the project. It should also give reasons for locating the project in a particular area to be given. To assist in the assessment of the environmental effects of a project every feasibility report must present the information on specific points, i.e., populations water land, air, flora, fauna, effects arising out of projects pollution and other environmental disruptions etc.

The report should contain a list of important items of capital equipment and also a list of the operational requirements of plant, water, power, personal, organizational structure envisaged, transport costs, activity wise phasing of construction and factors affecting it.

4. Marketing Plan: -

The feasibility report should contain the following items:

Data on the marketing plan. Demand and prospective supply in each of the areas to be served.

The methods and the data used for main estimates of domestic supply and selection of the market areas should be presented. It should present an analysis of past trends in prices.

5. Capital Requirements and Cost: - The estimates should be reasonably complete and properly estimated. Information on all items costs should be carefully collected and presented.

6. Operating Requirements and Costs: - Operating costs are essentially those costs which are incurred after the commencement of commercial production. Information about all items of operating costs should be collected. Operating costs relate to cost of raw materials and intermediates, fuel, utilities, labor, repair and maintenance, selling expenses and other expenses.

7. Financial Analysis: - A proforma balance sheet for the project data should be presented. Depreciation should be allowed for on the basis specified by the Bureau of Public Enterprises. Foreign exchange requirements should be cleared by the Department of Economic Affairs. The feasibility report should take into account income tax rebates for priority industries, incentives for backward areas accelerated depreciation etc. The sensitivity analysis should also be presented. The report must analyze the sensitivity of the rate of return of change in the level and pattern of product prices.

8. Economic Analysis: - Social profitability analysis needs some adjustment in the data relating to the costs and returns to the enterprise. One important type of adjustment involves a correction in input costs, to reflect the true value of foreign exchange, labor and capital. The enterprise should try to assess the impact of its operations on foreign trade. Indirect costs and benefits should also be included in the report. If they cannot be quantified they should be analyzed and their importance emphasized.

PLANNING COMMISSIONS'S GUIDELINES (CHECK LIST) FOR FEASIBILITY REPORT:-

1. Examination of public with respect to the industry.
2. Broad specifications of outputs and alternative techniques of production.
3. Listing and description of alternative locations.
4. Preliminary estimates of sales revenue, capital costs and operating costs of different alternatives.
5. Preliminary analysis of profitability for different alternatives.
6. Marketing analysis.
7. Specification of product pattern and product prices.
8. Raw material investigation and specification of sources of raw material supply.
9. Estimation of material energy, flow balance and input prices.
10. Listing of major equipment by type, size and cost.
11. Listing of auxiliary equipment by type, size and cost.
12. Specification of sources of supply for equipment and process know-how.
13. Specification of site and completion of necessary investigation.
14. Listing of buildings, structures and yard facilities by type size and cost.
15. Specification of supply sources connection costs and other costs for transportation services, water supply and power.
16. Preparation of layout.
17. Specification of skill-wise labor requirements and labor costs.
18. Estimation of working capital requirements.
19. Phasing of activities and expenditure during construction.
20. Analysis of profitability.
21. Determination of measures of combating environmental problems.
22. Analysis of the past performance of the enterprise responsible for implementing and running the project with respect to project completion, capacity utilization, profitability, etc.
23. State of preparedness to implement the project rapidly.

Economic Viability: – Above everything, a project should be viable. It should break even on a cash basis in the first 6 to 8 months. It should break even (on cost basis) in the first 9 to 10 months. One must get an accounting profit no matter how small it is, in the very first year.

He should not think he cannot earn profits in the first year. He can if he is determined.

He must declare

a dividend of at least eight percent in the second year. Many entrepreneurs deceive themselves about the economic viability after fooling the financial institutions.

An entrepreneur can least

afford to deceive himself of all people.

2.0 Technical analysis:

1. Material inputs & utilities:
 - Raw materials – agricultural, mineral, livestock, forest, marine products.
 - Processed industrial materials & Components
 - Utilities – Power, water, fuel, steam, air etc.
2. Manufacturing process / Technology
 - Choice of technology
 - Acquiring technology
 - Appropriateness of technology
3. Product mix
 - Market requirements
 - Variations in size, quality
 - Product, Price, Place & Promotion
4. Plant capacity
 - Technological requirement
 - Input constraints
 - Investment cost
 - Market conditions
 - Resources of the firm
 - Government policy
5. Location & site
 - Proximity to raw materials
 - Availability of infrastructure – water, Power, fuel, transport, communications etc.
 - Nearness to market
 - Government policies
 - Availability of labour and their attitudes
 - Climate, pollution, facilities like schools, entertainment etc.
6. Machinery & equipment
 - Selection, procurement, installation & commissioning
7. Structures & civil works
 - Site preparation
 - Buildings & structures
 - Outdoor works
8. Project charts & layouts
 - General functional layout
 - Material flow diagrams
 - Production line diagrams
 - Transport layout
 - Utility consumption layout
 - Communication layout
 - Organizational chart
 - Plant layout
9. Work schedule
 - Installation phase
 - Phasing of investment
 - Develop plan of operation

3.0 Financial Analysis:

1. Cost of Project:

- Land & site development
 - Buildings & civil works
 - Plant & machinery
 - Technical know-how & engg. Fees
 - Expenses of foreign technicians & training
 - Miscellaneous fixed assets
 - Preliminary & capital issue expenses
 - Pre-operative expenses
 - Provision for contingencies
 - Margin money for working capital
 - Initial cash losses
2. Means of finance:
- Share capital
 - Term loans
 - Debenture capital
 - Deferred capital
 - Incentive sources
 - Miscellaneous sources
 - Planning the means of finance:
 - Norms of regulatory bodies and financial institutes
 - Key business considerations like cost of capital, risk, control, flexibility etc.
3. Estimates of sales & Production:
- Capacity utilization
 - 40-50% in 1 year
 - 50-80% in 2 year
 - 80-90% in subsequent years
 - Selling price is realizable value
 - Production & sales assumed to be equal
 - Changes in selling price may be matched with changes in cost of production
4. Cost of Production:
- Material cost, labor cost, utilities cost, factory overheads
5. Working capital requirement & financing:
- Raw materials & components
 - Work in process
 - Finished goods stock
 - Debtors
 - Operating expenses
 - Sources of WC
 - Advances by commercial banks
 - Trade credit
 - Accruals and provisions
 - Long term sources of financing
 - 25% of current assets must be supported by long term sources of financing i.e. margin money
6. Profitability projections: (Estimates of working results)
- | | |
|----------------------------------|--|
| A: Cost of production | J: Operating profit: G-H-I |
| B: Total administrative expenses | K: Other income |
| C: Total sales expenses | L: Preliminary expenses
written off |
| D: Royalty & know-how | M: Profit / loss before taxes: |

E: Total cost of prodn. (A+B+C+D)	J+K-L
F: Expected sales	N: Provision for tax
G: Gross profit before interest	O: Profit after tax: M-N
H: Total finance expenses	Less dividend
I: Depreciation	- Preference capital
	- Equity capital
	R: Retained profit
	Q: Net cash accrual : P+I+L

7. Breakeven analysis:

Break even Point in units = $\frac{\text{fixed costs}}{(\text{unit selling price} - \text{unit variable cost})}$

$$= \frac{\text{Fixed cost}}{\text{Contribution}} \times \text{Expected prodn. in nos. Nos.}$$

BEP (% of installed capacity) = $\left(\frac{\text{Fixed cost}}{\text{contribution}} \right) \times \text{Expected Capacity utilization in the year.}$

$$\text{BEP(in Rs.)} = \frac{\text{Fixed cost}}{\text{Contribution}} \times \text{Expected sales realization in the year}$$

Contribution = sales realization – variable cost

8. Projected cash flow statements:

Cash flow statement shows the movement of cash into and out of the firm and is net impact on the cash balance with the firm.

Sources of funds Disposition of funds

9. Projected balance sheet:

Balance sheet shows the balances in various assets and liabilities.

It reflects the financial condition of the firm at a given point of time.

Liabilities	Assets
Share capital	Fixed assets
Reserves & surplus	Investments
Secured loans	Current assets, loans, advances
Unsecured loans	Miscellaneous Expenditure & losses
Current liabilities & provisions	

4.0 Project financing:

It refers to the means of finance employed for meeting the cost of the project. The long-term sources used for meeting the cost of the project are known as means of finance.

1. Equity (Owned funds)

- Ordinary shares
- Preference shares

2. Debt (barrowed funds)

- Secured from financial institutes
- Debentures – convertible, non-convertible
- Public deposits
- Rupee term loans
- Foreign currency loans
- Euro issues – Global depository receipts , Euro convertible bonds
- Deferred credit

3. Lease financing

- Maintenance lease
- Financial lease

- Operating lease
- Net lease

4.1 Classification of capital:

1. Fixed capital: Funds required for acquiring fixed assets.
2. Working capital: Funds required for operations and includes raw materials; work in processes, finished goods, wages and salaries etc.
3. Venture capital: Venture capital is thought of as a creative capital, which is expected to perform economic functions different from other investment vehicles, which primarily serve as expansion capital.
It is the equity support to fund new concepts that involves a high risk and at the same time has high growth and profit potential.

Institutions providing venture capital:

- The technology Development and Investment corporation of India (a subsidy of ICICI)
- Technology Development fund set up by IDBI
- The Equity development Scheme - SBI capital markets Ltd., CANFINA.
- India Investment Fund. – Grindley's bank.

4. Seed capital

It is the capital to be subscribed by the promoters as required by the financial institutions. Financial institutions through seed capital assistance supplement the resources of promoters of the small and medium scale industries.

1. Special seed capital assistance scheme: Rs. 2.0 lakhs or 20% of the project cost whichever is less. SFCs
2. Seed capital assistance for projects costing not more than Rs. 200 lakhs – Rs. 15 lakhs max. – by IDBI
3. Risk capital foundation scheme for projects of Rs. 150 to Rs. 200 lakhs – Rs. 15 to 40 lakhs - by IFCI.

4.2 Financial institutions:

1. Industrial finance Corporation of India (IFCI)
2. The Industrial Development Bank of India (IDBI)
3. The Industrial Credit and Investment corporation of India (ICICI)
4. The National Bank for Agriculture and Rural Development (NABARD)
5. The Small Industries Development Bank of India (SIDBI)
6. Industrial Investment bank of India (IIBI)
7. Life Insurance Corporation of India (LIC)
8. General Insurance Corporation of India (GIC)
9. Export Import bank of India (Exim Bank)
10. Khadi & Village Industries commission (KVIC)
11. National Small Industries Corporation Ltd. (NSIC)
12. State industrial Development Corporations (SIDCs)
13. State Small industries Development Corporations (SSIDCs)
14. State Financial Corporations (SFCs)
15. Commercial banks

4.3 Institutions engaged in entrepreneurial development:

1. Small Industries Extension Training Institute, Hyderabad (SIET)
2. Small Industries Service Institute (SISI)
3. Small Industries Development Organization (SIDO)
4. Entrepreneurial Development Institute of India (EDII) set up by IFCI

5. National Institute for Entrepreneurship and Small business Development (NIESBUD)
6. Gujarat Industrial and Investment Corporation (GIIC)
7. Indian Investment Center (IIC)
8. Entrepreneurial Motivation Training Center EMTC)
9. Xavier Institute of Social Sciences Ranchi.
10. Center for Entrepreneurship Development , Ahmadabad (CED)
11. Rural entrepreneurship development (RED) institute.
12. National science and technology Entrepreneurship development Board (NSTEDB)
13. Rural Management and management centers at Maharashtra and Training cum Development centers RMEDC)
14. Management Development Institute (MDI)

Unit-IV

Project Management during construction phase, project organization, project planning and control using CPM, PERT techniques. Human aspects of project management. Assessment of tax burden.

Typically, a firm is organised into various departments – production, purchasing, marketing, finance, personnel, engineering and research and development. Some of these have line function and the others a staff function.

Line managers have the principal responsibility for achieving the goals of the firm and are vested with decision making authority. Staff managers primarily serve in an advisory capacity and within the staff departments, they enjoy administrative powers. Traditional form of organisation is not suitable for project management. The reasons are:

- i) A project is a non-routine, non-repetitive undertaking often plagued with many uncertainties.
- ii) The relationships in a project setting are dynamic, temporary and flexible, and
- iii) A project requires coordination of the efforts of persons drawn from different functional areas and contributions of external agencies.

Project work calls for design, development, procurement, construction and commissioning work.

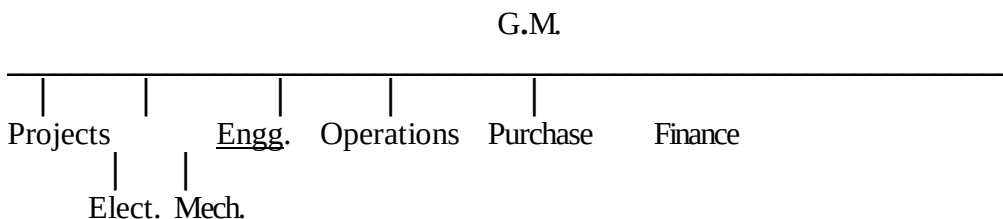
Project management consists of

- I. Forms of project organisation
- II. Project planning
- III. Project control
- IV. Human aspects of project management
- V. Pre-requisites for successful project implementation

I. **Forms of project organisation**: - The responsibilities of a Project Manager or Project Coordinator are: integrating the activities and functions of various departments and external organizations involved in the project work.

Depending upon the authority given to the project manager the following one of the three types of organisation take place:

- i) Line and staff organisation: - This is suitable for effective realization of project objectives and effectively utilises resources. It can be best suitable for small projects.
- ii) Divisional organisation: - In this form of organisation, unnecessary duplication of specialists in the company, because of the necessity to allocate them in total to each project. In this it is not possible to utilize resources effectively but can effectively realize the project objectives.
- iii) Matrix organisation: - This type of organisation realises both project objectives and efficient utilises the resources. Personnel working in project have dual responsibility to their functional superior as well as the project manager. The authority is shared between the both. It considerably improves the prospects fulfilling the time and budget targets.



Matrix Organisation

II. **Project Planning**: — without effective planning there may be chaos.

i) Functions of planning:—

— organising the work and allocating responsibilities to individuals

- it is a means of communication and coordination between all those involved in project
- induces people to look ahead
- instills a sense of urgency and time consciousness
- establishes the basis for monitoring and control

ii) Areas of Planning:—

- planning the project work - activities must be spelt out in detail. They should be properly scheduled and sequenced.
 - planning the manpower and organisation (managers, technologists, operators and others) must be estimated and the responsibility allocates.
 - planning the money — the expenditure of money in a time—phased manner must be budgeted.
 - planning the information system — for monitoring the project (information required) must be defined.
3. Project objectives: — well defined objectives and policies serve the decisions to be made by the Project Manager. A clear articulation of the priorities of management will enable the project manager to take expeditious actions.
4. Work breakdown structure: — it helps in
- effective planning by dividing the work into manageable elements which can be planned, budgeted and controlled
 - assigning the responsibility to project personnel and outside agencies.
 - development of control and information system

‘Work breakdown and project organisation’: the work breakdown structure has to be integrated with the project organisation. This calls for blending the vertical breakdown of the work (as arrived in the work breakdown structure) with the project organisation structure. This results in delineation of project tasks which are the specific responsibilities of organizational units/managers. The technical make given to the project task is ‘cost account’. A cost account represents a unit of work i) which is defined in fairly concrete terms, ii) for which a single person is responsible and iii) for which a budget of expenditure and manpower requirements can be prepared meaningfully.

5. Life cycle of Project: — important stages are

- project development and preliminary engineering
- bidding and contract negotiations
- engineering design
- purchase and procurement
- construction
- commissioning

6. Tools of planning: — the oldest formal planning tool is the bar chart or multiple activity chart. In the last four decades, network techniques have received considerable attention.

Bar Chart: — this is a pictorial device in which the activities are represented by horizontal bars on the time axis. The left hand bar shows the beginning time, the right hand end the ending time. The duration of the activity is indicated by the length of the bar. The manpower required for the activity is shown by a number on the bar.

An illustrative bar chart

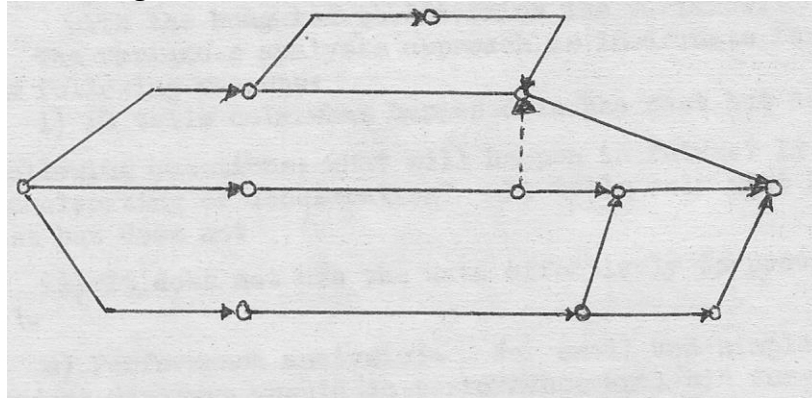
	Time in weeks from project start				
	10	20	30	40	50
Design	7 └───┘				
Purchase of parts		2 └───┘			
Fabrication			16 └───┘		
Assembly				4 └───┘	

The advantages of the bar chart are: i) it is simple to understand, ii) it can be used to show progress, and iii) it can be

used or manpower planning. The bar chart, however, suffers from some disadvantages which limit its usefulness: i) it cannot show interrelationship among activities on large, complex projects, ii) there may be a physical limit to the size of the bar chart, which may limit the size of the project that can be planned with this technique, and iii) it cannot easily cope with frequent changes or updating

Network Techniques: — these are more sophisticated than the traditional bar chart. In these techniques, the inter—relationships are represented by an arrow diagram.

An illustrative Network Diagram



The advantages of network techniques are: i) they can effectively handle inter—relationships among project activities, ii) they identify the activities which are critical to the completion of the project on time and indicate the float (or spare time) for other activities, iii) they can handle very large and complex projects, and iv) they can be easily computerized and updated.

While the network techniques are a superior tool for project planning, they suffer from several drawbacks: i) being more complicated than the traditional bar chart they are not easily understood by the project personnel, and ii) they do not define an operational schedule which tells who does what and when.

III. Project control: — it concerns a regular compliance of performance against targets, a search for the causes of deviation and a commitment to check adverse variance. It serves two major functions: i) it ensures regular monitoring of performance and ii) it motivates project personnel to strive for achieving project objectives.

Reasons for ineffective control are:

a) Characteristics of the Project:

— Keeping track of physical performance and expenditure on hundreds or thousands (or tens of thousands) of activities which are often non—routine is a stupendous task.

— Coordination and communication problems multiply when several organizations are involved in the project.

b) People Problems: — Most of the operational managers, used to the

steady rhythm of normal operations and routine work lack the experience, training, competence, and inclination to control projects.

c) Poor Control and Information System: Some of the weaknesses observed in the control and information system are:

delay in reporting performance

— Inappropriate level of detail

— unreliable information

d) Variance analysis approach:— The traditional approach to project control involves a comparison of the actual cost with the budgeted to determine the variance.

The variances analysis approach is inadequate for project control for the following reasons:

i) It tells only what happened in the past but does not answer the following questions: What will happen in future? Is the rate of work accelerating or decelerating? It tells only what happened in the past

ii) It does not use the data effectively to provide integrated control.

e) Performance analysis: — for small and simple projects, the project managers would do performance analysis for

the project as a whole, or for its major components. As the project becomes larger and more complex, performance analysis needs to be done for individual segments of the projects which are referred to as 'cost accounts'.

As we have seen earlier, a 'cost account' is illogical management centre. Performance analysis is a modern approach to control.

IV. Human aspects of project management: — Project manager must successfully handle problems and challenges relating to

- i) Authority
- ii) Orientation
- iii) Motivation
- iv) Group functioning

i) Authority: The project manager has to explain the logic and rationale for the project activities, show receptivity to the suggestions made by others, avoid unilateral imposition of decisions, eschew dogmatic postures, and search for areas of agreement which can be the basis of acceptable solutions.

ii) Orientation: — Most of the managers working for a project are usually engineers (or technologists). Typically, an engineer

- works with physical laws, characterised by mathematical precision, as his tools
- Adopts a structured, mechanical approach to his problems
- seeks an enduring solution to his problems
- attaches a high value on technical perfection

But an engineer performs the tasks of planning, organising, directing, and, controlling the resources of the firm in a world of uncertainty. Adopts a more creative approach to solve non—programmed and unstructured problems.

Attaches greater importance to efficient utilization of resources and resolution of human relation problems

Thus the project manager has to strengthen the managerial orientation of project personnel so that the project goals and objectives can be efficiently achieved within the constraints of time and budget.

iii) Motivation: — the principal behavioral factor which the project manager can influence is the motivation of the project personnel. Humans are motivated by a variety of needs: physiological needs, social needs, recognition needs, and self—actualization needs. The traditional approach to management was based on the assumption that human beings regard work as unpleasant, shirk responsibility, and ordinarily employ inefficient and wasteful methods. A great deal of pressure has to be applied to overcome this. Behavioural research, however, has shown that while some pressure is beneficial, an excess of it is undesirable. Beyond a certain point, pressure is dysfunctional.

iv) Group functioning: — in a large complex project, many persons drawn from different functions, departments, and organizations are involved. This leads to formation of groups, formal and informal. These are of three types:

- Vertical groups consist of people drawn from different levels in the same department, or function, or company.
 - Horizontal group consists of people drawn from different functions, departments, and companies, but occupying similar hierarchical positions.
 - Mixed group consists of people drawn from different levels from various functions, departments, and companies.
- Due to vertical group formation in the organisation conflicts may arise based on "we/they". The horizontal group serves as channels of communication, by their influence; they can strengthen the commitment to the project. The mixed group tends to promote greater cohesion of the total project organisation. It is very conducive in creating a project attitude and developing an overriding commitment for the project. Hence the project manager should strive to establish a mixed group as the principal group of the project. However, it is difficult to establish such a group because of the temporary nature of a project--when members of a group know that the group would be dissolved sooner or later, they retain strong links with their parent company or department.

'Building Effective Groups':— an effective group consists of members who are satisfied and committed and who strive for the attainment of project objectives, without dissipating their energies in inter—personal and inter—group conflicts. The manifest signs of an effective group are: pride in the project, supportive behavior, coordinated endeavor, mutual respect, and resilience during trying periods. An ineffective group, on the other hand, consists of disgruntled members who are more involved in inter—personal and inter—group rivalries and less concerned about project goals. Such a group is characterised by apathy, animosity, mutual bickering, disjointed efforts, cynical attitudes, and low morale.

For building an effective group, firm must pursue a genuinely participative style of management. With this managerial philosophy, the project manager can facilitate the development of mutual trust and acceptance, open communication, cooperation, and project attitude. In this task, he needs leadership capabilities, sensitivity to human nature, perceptiveness, concern for welfare of their maturity, and impartial approach. Clearly this is a difficult and challenging task.

V. Prerequisites for successful Project implementation: — Time and cost over—runs of projects are very common in India, particularly in the public sector. The Annual Report of the Ministry of Programme Implementation for recent year released that 65% of the projects have suffered time over runs, which have gone as high as about 200% — the average delay in commissioning these projects was about 3 years and about 68% of the Central government projects, have suffered cost over—runs, which have been as high as 75%. A similar analysis of state projects would perhaps reveal an even more dismal picture.

Due to such time and cost over—runs projects tend to uneconomical, resources are not available to support other projects, and economic development is adversely affected.

Some important things to be done to achieve successful completion of projects are:—

i) Adequate formulation: — Care must be taken to avoid superficial field investigation, cursory assessment of input requirements, slipshod methods used for estimating costs and benefits, omission of project linkages, flawed judgments because of lack of experience and expertise, undue hurry to get started and deliberate over-estimation of benefits and under— estimation of costs so that the appraisal and formulation of the project is thorough, adequate, and meaningful.

ii) Sound project organisation: — The characteristics of a sound project organisation are: it is led by a competent leader who is accountable for the project performance, the authority of the project leader and his team is commensurate with their responsibility, adequate attention is paid to the human side of the project, systems and methods are clearly defined and rewards and penalties to individuals are related to performance.

iii) Proper Implementation Planning:— Implementation planning before commencement of actual implementation should plan for a comprehensive time plan for various activities like land acquisition, tender evaluation, recruitment of personnel, construction of buildings, erection of plant, arrangement of utilities, trial production run etc., estimate meticulously the resource requirements (manpower, materials, money etc) for each period to realise the time plan, define properly the inter-linkages between various activities of the project and specify cost standards.

iv) Advance Action: - Advance action on the following activities may be initiated: acquisition of land, securing essential clearances, identifying technical collaborators/consultants, arranging for infrastructure facilities, preliminary design and engineering and calling tenders.

To initiate advance action with respect the above some investment is required. Clearly, if the project is not approved, this investment would represent an anfractuous outlay. However, the substantial savings (in time and, cost) that are expected to occur, should the project be approved (a very likely event, given the prima facie desirability of the project) often amply justify the incurrence of such costs.

v) Timely availability of funds: - Firms which have a comfortable liquidity position are in general, able to implement Projects expeditiously and economically. Such firms can initiate advance actions vigorously, negotiate with suppliers and contractors aggressively, organise input supplies quickly, take advantages of opportunities to effect economies, support suppliers in resolving their problems so that they can in turn rebound to the successful completion of projects, and sustain the morale of project-related personnel at a high level.

So, once a project is approved, adequate funds must be made available to meet its requirements as per the plan if implementation--it would be highly desirable if funds are provided even before the final approval to initiate advance action.

vi) Judicious equipment tendering and procurement: - Over reliance on indigenous suppliers may mean delays and higher uncertainty about the technical performance of the project. It is advantageous from the point of view of cost and time regarding foreign suppliers through global tendering but foreign exchange which is very valuable to the country flows out. Hence a judicious balance must be sought which moderates the outflow of foreign exchange and provides reasonable fillip to the development of indigenous technology.

vii) Better Contract Managements: - Help should be extended to contractors and suppliers when they have genuine problems—they should be regarded as partners in a common pursuit. Project authorities must retain latitude to off-load contracts (partially or wholly) to other parties well in time where delays are anticipated.

viii) Effective Monitoring: - In order to keep a tab on the progress of the project, a system of monitoring must be established. This helps in anticipating deviations from the implementation plan, analysing emerging problems, taking corrective action,

NETWORK TECHNIQUES FOR PROJECT MANAGEMENT:-

Financial institutions and Govt. of India insist that network plan should accompany feasibility reports. There are two basic network techniques: PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method).

PROJECT SELECTION

Project ideas thus identified are translated into project schemes/ profiles by further screening them on the basis of their technical, economic and financial soundness.

The project scheme/profile comprises the following in brief:

1. Economic size
2. Status of Industry or Scope
3. Raw Material availability
4. Cost of production
5. Capital costs
6. Utility requirements
7. Infrastructure facilities needed
8. Profitability
9. Government Policy

A SS1 unit (Small Scale Industrial unit) generally enters the market which is localized, and already has existing manufacturers. The success of the new unit in competing with the existing manufacturers would depend upon a combination of its capability to identify and approach its targeted customer group and its marketing features like price, quality, delivery schedule, sales promotion, etc.

Criteria for Selecting a Particular Project:

After gathering a large number of project profiles, the entrepreneur should consider the following criteria for selecting a particular project:

PROJECT APPRAISAL FORMAT

Criteria	Project I	Project II	Project III	ProjectIV
1. Investment size				
2. Location				
3. Technology				
4. Equipment				
5. Marketing				
6. Power & Water				
7. Others' performance				
8. Working capital needs				
9. Labor component				
10. Economic viability				
Total				

1. Investment size: Professional managers who have worked in multi— national companies or large Indian companies, should think of starting medium—sized or large—sized units only. The investment size (project cost)

should be at least Rs.3 to 5 crores. They should not commit the common mistake of restricting the project size to less than Rs.2 crores so that they need not go to the all—Indian financial institutions. In fact, under the present circumstances, it will be much easier to get projects cleared by the all—India institutions, requiring even lesser promoter's contribution.

2. Location: A new entrepreneur should locate his project to the extent possible, in and around a State headquarters. There are many backward areas around such cities. It is necessary to have such a location to attract competent managers. This will also facilitate liaison with the State Electricity Board, State Industrial Development Corporation and various other agencies.

3. Technology: The first project should not be for product which requires high technology, necessitating foreign technical collaboration. It is better to go in for a product with proven technology that is indigenously available. It makes life easier to begin with.

4. Equipment: The entrepreneur should select the best equipment based on the advice of experienced technical consultants. He should not compromise on the quality of the equipment. Many entrepreneurs enter into some sort of a deal with the equipment manufacturers for a “kick—back” and in the process sacrifices quality. Industry is a goose that lays golden eggs. One should not be short—sighted and kill the goose by going in for poor quality equipment.

5. Marketing: It is not advisable to get into a project particularly the first, which would mean survival amidst cut—throat dog—eat—dog competition involving direct selling to the ultimate consumer. One should go in for products with a limited number (say 10 to 20) of industrial customers.

6. Power & Water: One should ensure abundant supply of these two inputs. Power intensive and water intensive projects should be avoided.

7. Others' performance: How are others doing? A new entrepreneur should explore as to how those already in the field are doing. He should not get into areas where seasoned industrialists fear to tread. As a rule, he should get into a line in which others are doing reasonably well.

8. Working Capital requirements: It is not desirable to choose a project, which is working capital intensive and whose cash (operating cycle is too long. The lending policies of banks are rather unpredictable these days. An entrepreneur is likely to run into trouble on this front if he underestimates his working capital requirements initially (to keep his margin money low). His problems will be compounded, if he has to buy from any government source (by paying advance) or to sell to any government agencies (which are well known for their delayed settlement of bills).

9. Labor component: The project should not be labor intensive, at least in the first instance. Labor productivity in India is very low and Indian labor is ultimately very costly, contrary to popular belief. A shrewd entrepreneur will avoid unskilled and semi—skilled labor and minimise the material—handling labor with proper buy—out policies and use of better equipment.

10. Economic viability: Above everything, a project should be viable. It should break even on a cash basis in the first 6 — 8 months. It should break even (on cost basis) in the first 9 — 10 months. One must get an accounting profit no matter how small it is in the very first year. He should not think he cannot earn profits in the first year. He can if he is determined. He must declare a dividend of at least eight percent in the second year. Many entrepreneurs deceive themselves about the economic viability after fooling the financial institutions. An entrepreneur can least afford to deceive himself of all people.

Overall evaluation: He should evaluate all the projects on hand on a multi—point scale as indicated in the statement. The project which scores maximum points should be selected for further exploration.

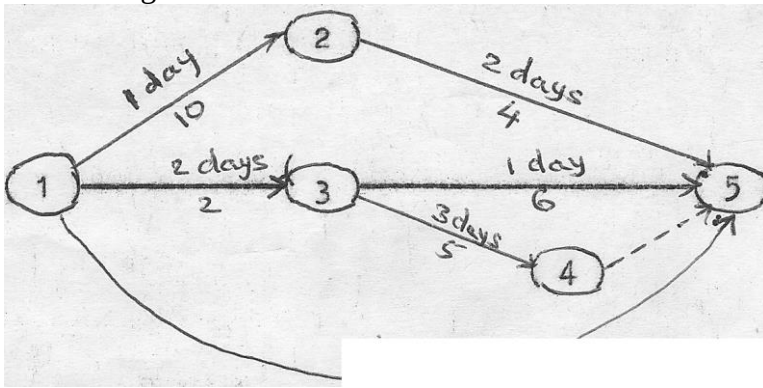
Further exploration: After short-listing the projects on the lines suggested above, he should do further exploration by consulting people in that line, meeting technical consultants and discussing with financial institutions. If he is satisfied in the second round also, then he should commission a detailed project report, by entrusting the job to some well—known consultants in the field. The maxims “Look before you leap” and “Think before you act” will save a new entrepreneur time and money.

Meeting manpower requirement:-

The manpower required for the activities is indicated by the length of the bar in the bar chart (most suitable for small projects). These bar chart can be used for manpower planning. For large and complex projects Network techniques can be used.

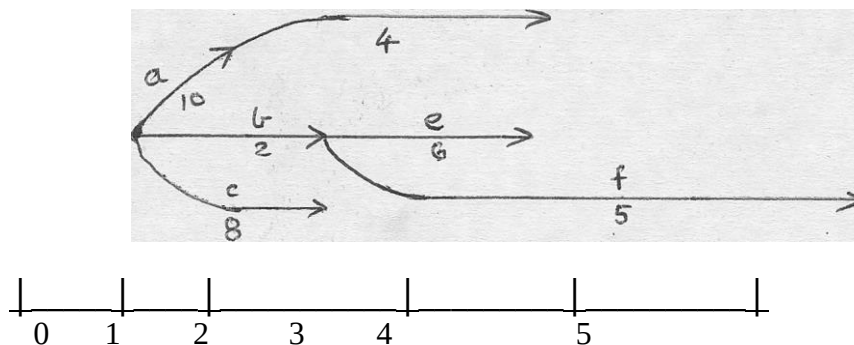
The manpower requirement is shown below the activity arrow. Let us consider a small project for which the

network diagram in shown in below:-



Looking at the manpower requirement for the early start schedule we find that it is 20 for the first day, 14 for the second day, 15 for the third day, 5 for the fourth day, and 5 for the fifth day. Real life situations, however, there may be restrictions on the availability of resources. For example, manpower supply may be limited. When restrictions exist various schedules may have to be considered to find out which one is most appropriate in the light of these restrictions.

In the above network diagram, only 12 men are available for the project (manpower resource constraint). The early start schedule of this project is shown as a graph on the horizontal time scale.



Looking at the manpower requirement for the early start schedule we find that it is 20 for the first day, 14 for the second day, 15 for the third day, 5 for the fourth day, and 5 for the fifth day. Obviously, this schedule is unacceptable in view of the manpower constraint.

So, we explore the possibility of shifting activities. Our efforts at shifting activities, keeping the project duration at five days, soon reveals that no schedule is feasible with only 12 men. So we extend the duration of the project, by one day and try various schedules to see whether we can find a feasible schedule. A little juggling of activities shows that a schedule like the one shown as below is feasible -- this is the best we can do.

GENAERAL FORMAT ON SCOPE OF FEASIBILITY REPORT

1. Introduction
2. Summary and Recommendations
3. Product
 - a) Capacity
 - b) Chemistry of the product
 - c) Specifications
 - d) Properties
 - e) Applications and uses
4. Market Potential
 - A) Existing installed capacity and actual production for the last five years.
 - B) New capacities under consideration/implementation.
 - C) Demand pattern with specific reference to requirements in the southern region based on desk research and field survey.
5. Process and know-how
 - a) Description of different processes available
 - b) Selection of the process for this project and the reasons for the same.
 - c) Selection of suitable know-how for the said project based on technical and economic evaluation from available data.

Prospective collaborator and terms of collaboration.
- 6* Plant and Machinery
 - a) List of imported/indigenous machinery
 - b) Broad specifications of the machinery.
 - c) List of manufacturers of the machinery.
 - d) Critical items of the machinery.
7. Location

Actual site selection survey -Availability of raw material, electricity, water and other infrastructure facilities are to be considered and a suitable location be decided.
8. Plot Plan and Building

Overall plot plan along with building plan.
9. Raw Materials
 - a) Raw material requirements and their specifications.
 - b) Availability and sources of the raw materials (imported and indigenous).
10. Utilities

Requirements of power, water others including fuel, steam, Inert gas, etc., and their specifications.
11. Effluents
 - a) Nature and type of effluent/specifications and quantity of effluent. B) Effect of effluent if discharged as such.
 - c) Effluent treatment suggested.
12. Personnel Requirement
 - a) Manpower requirement. (top, middle and lower level).
 - b) Organization structure.
13. Capital cost
 - a) Project cost giving break-up details (based on budgetary estimates). B) Margin money for working capital in the project cost
14. Working Capital

Details of requirements of working capital & norms.
15. Mode of Finance

Based on the debt equity ratio and other financing norms followed by financing institutions.

16. Manufacturing cost
A) Unit cost of production.
B) Projected cost of production for 10 years.
17. Financial Analysis
a) Profitability statement for 10-years
b) Break-even and sensitivity analysis.
c) Pay back period.
d) Budgetary balance sheet for 10 years

18. Implementation Schedule

Time schedule for the implementation of the project will be given along with a Bar Chart/Network diagram.

LIST OF APPLICATION OF FORMS:-

The following is the list of forms to be submitted along with the application for financial assistance.

- I Letter Addressed to the Bankers
- II Existing Long-term Borrowings
- III Existing Short-term Borrowings
- IV Distribution of Shareholdings
- V Particulars of Buildings
- VI Particulars of Imported Machinery
- VII Particulars of Indigenous Machinery
- VIII Raw Materials Requirements
- IX Estimates of the Cost of Project
- IXA Calculation of Contingency
- IXB Calculation of Margin Money
- X Means of Financing
- XA Proposal for Raising Share Capital
- XB Sources of Expenditure Incurred
- XI Estimates of Cost of Production
- XII Estimates of Working, Results
- XIIA Estimates of Production and Sales
- XIIB Calculation of Wages & Salaries
- XIII Unit Cost of Production
- XIV Cash Flow Statement
- XV Projected Balance Sheet

Government of Andhra Pradesh - New Industrial Policy 1992

The Government of Andhra Pradesh announced its new Industrial Policy 1992 to accelerate Industrial Development in the State within the broad parameters of the policies laid down by the Government of India.

Andhra Pradesh occupies a strategic position as the Gateway of the South, centrally situated at the very centre of our sub-continent. The Eastern Port of Visakhapatnam provides an outlet to the hinterland of not only our state but also Madhya Pradesh and Orissa. This natural advantage is supplemented by resource abundance of minerals. With long Coast line abundant Aqua-marine wealth is another added attraction. There is no dearth of trained labor, skilled manpower or managerial talents in the State. The State can boast of the very best industrial peace. The State has unlimited potential for 100% export oriented units in the field of polished granite, artificial granite, finished leather goods, processed and preserved meat and other foodstuffs including fruit pulp as well as artificial jewellery, precious and semi-precious stones.

Ours is also the largest manufacturer and exporter of bulk drugs, basic drugs and pharmaceutical formulations. The State has huge deposits of the Bauxite nearly 450 million tons in and around Chintapally taluq of Visakhapatnam district; 90% of India's Barytes and nearly 60% of Worlds' Barytes is in the Cuddapah district; the State has the largest and the longest, contiguous and concentrated 'Coal and Limestone' deposits in

the World. The State has enormous potential for Agro-based industries. The State has nearly one lakh Small Scale Units/Tiny units providing employment to over 8 lakh persons. Small Scale Industry generates much more employment than Medium and Large Scale units for a particular level of investment. Andhra Pradesh registered a growth rate of 2.7% during the Seventh five year plan period with a manufacturing sectors' share of 11.42 in the net State Domestic Product in 1990-91 against 11.09% in 1989-90.

The Government of India announced a package of incentives for the development of Small Scale and Tiny industries, which contribute more than 50% of the employment and more than 50% of our exports. The new National Trade Policy liberalized and simplified the procedures to further encourage export oriented industries, discourage indiscriminate and unjustified imports of Capital goods, intermediaries and raw materials, In a, way licensing has been done away with, except for a few core sector items. In order to maximise the benefits of the New Industrial Policy of the nation* the State Government has brought out revised guidelines to ensure smoother registration of industries in the SSI/Tiny sector.

Incentives:-

a) Investment subsidy: The scheme allows an investment subsidy of 20% of the fixed capital investment subject to a maximum of Rs.20 lakhs for new industrial units being set up in Srikakulam, Anantapur and Adilabad districts. In all other districts of the State excluding those being located within the municipal limits of Hyderabad, Vizag and Vijayawada, an investment subsidy 15% of the fixed capital investment subject to a maximum of Rs.15 lakhs is allowed.

b) Sales Tax Incentive:

i) Sales Tax deferment for Large and Medium Scale Industries: New Units being set, up in Srikakulam, Anantapur and Adilabad are allowed Sales Tax deferment up to a maximum of 100% of the fixed capital investment in a period of 10 years. The Sales deferment up to a maximum of 75% of fixed investment in a period of 10 years for the units set up in Medak, Warangal, Khamam, Karimnagar, Nizamabad, Nalgonda, Mahaboobnagar, Ongole, Nellore, Kurnool, Cuddapah, Chittoor and Vizianagaram.

The sale tax deferment up to a maximum 50% of the fixed capital investment in a period of 10 years for the units in Hyderabad (excluding Municipal Corporation Area), Ranga Reddy, Krishna (excluding Municipal Corporation Area), Guntur, East Godavari, West Godavari and Vizag (excluding Vizag Municipal Corporation Area).

ii) Sales Tax exemption for Small Scale Industries: New units coming up anywhere in the State (excluding Hyderabad, Vizag and Vijayawada Municipal Corporation Area) are exempted from Sales Tax up to a maximum of 100% fixed capital investment or Rs.35 lakhs whichever less in a period of 5 years.

Note: The above incentive will not be available for 75 specified industries. This scheme has been announced to be in

force for a period of 3 years initially i.e., up to Oct.2,1992. The State Government proposed to continue the scheme through up to the end of Eighth Plan i.e., up to 31-3-1997.

iii) Rebate on Power Charges: Now Industrial Units, barring certain common or specified industries and power-intensive industries, are allowed a rebate of 25% in the power bills of consumption of energy and demand for a period of 3 years from the date of commencement of production.

iv) Sales Tax incentive and reliefs:

The State Government have taken cognisance of certain difficulties faced by industry vis-à-vis the concessions available in certain neighboring States etc., and have already extended the following incentives and reliefs:

— Rate of tax on electronic goods reduced to 2% only

— Reduction of % of Sales Tax by nearly half i.e. 50% in case of mini-Cement Plants

- Sales Tax on wheat and wheat products has been exempted

- In order to see that the local industry Pays a Sales Tax of 4% in effect for supplies made to State and Central Government Departments, Public Sector Undertakings and Local bodies, the State Govt. has set up a revolving fund to cover the purchases made by four public sector undertakings namely APSRTC, APSEB, Singareni Colleries and APDDC. Further procedure to extend this benefit for purchases being made by other Govt. undertakings etc.

— The State Govt. have simplified the section 5-B of APGST Act to enable the manufacturing units obtaining their inputs, namely raw materials, component. consumables and packing materials etc. at concessional rates of tax 4% by introducing registration of manufacturing units.

Special Emphasis for developing SSI and Tiny sector:

a) The Govt. of India have formulated a new scheme called "Integrated Infrastructural Development (including Technological Back-up Services) for Small Scale Industries in Rural Backward Areas". The expenditure being borne in the ratio of 1:1:3 between the Central Govt., State Govt., and Financial Institutes. The State Govt. will provide suitable land of the requisite size endowed with infrastructure. While choosing the location of the Centre, the State Govt. will keep in mind that workers engaged do not get displaced from their traditional dwelling places and that they do not have to walk or bicycle more than 5 kilometers to go to their place of work.

b) Registration: As the first step, a procedure of registration of the units has been made very simple. The certificate of provisional registration has been ordered to be issued across the desk and in any case not later than a week. The permanent registration certificate will be given positively within a month at the latest.

a) Marketing Assistance: The major draw back for smooth functioning of this sector is lack of sustained marketing assistance, apart from non-availability of raw material and of easy credit in many cases. The State Govt. already introduced Marketing assistance scheme implemented through APSSIDC (Andhra Pradesh Small Scale Industrial Development Corporation). The scheme stipulates that all Govt.

departments should draw their requirements of 12 items from the local SSI/Tiny sectors only through APSSIDC.

d) Presently 75 Product lines covered mostly in Tiny Sector are exempted from all types of clearances. Govt. proposes to extend this list, to cover 23 more product lines.

Liberalization on avoidable "over heads" in setting up an industry:

The State Govt. is considering the following items:

- a) Automatic exemption from Urban Land Ceiling Act for industry set up in recognized industrial estates/areas.
- b) Treating each partner of a partnership firm as an individual for the determination of land holding under Land Ceiling Act.
- c) Informing the Revenue authorities shall suffice for conversion of agricultural land to non-agricultural purpose for setting up an industry.
- d) Authorising the Collectors concerned to hand over the lands required for industrial purpose, whether by private individuals/entrepreneurs/promoters or by corporate bodies in favor of AP11C for, in turn, will evaluate the land, allot the same as per procedure, and report to Govt.
- e) To delegate powers to accord clearances under Town Planning Regulations to District level in order to speedup implementation of projects.

ANDHRA PRADESH - INVESTMENT FRIENDLY

Andhra Pradesh the hub of industrial activity in South India, is going all out to welcome and encourage industrial entrepreneurs. A combination of inherent strengths, dynamic policies and a quick responsive Govt. makes A.P. the ideal location for industry.

- * Number four ranking in investment in existing units.
- * Stands fifth in cumulative disbursement of assistance by Central Financial Institutions up to March, 1992.
- * 208 industrial development areas and industrial estates over 10,000 acres with all basic facilities.
- * A software park for electronics industry.

FINANCE:

Down the ages, A.P. has always been a rich State.

- * There are presently 159 large and medium scale industries with an investment of Rs.22,676 crores in A.P.
- * A.P. has one of the highest per capita household consumer expenditure
- * The state ranks Fourth in state domestic product

- * The state has a policy to participate in the equity of new industrial ventures

INFRASTRUCTURE:

A strong support system

- * 370 acres of land and ready buildings in the Visakhapatnam Export Processing Zone
- * 800 acres in Visakhapatnam Port-based Exim Park
- * 800 acres in the vicinity of Kakinada Port
- * A Chemical Complex with a common effluent treatment plant at 50 Kilometers from Hyderabad
- * A software park for electronic industry in Hyderabad
- * The State Govt. will welcome private initiative for setting up infrastructural facilities for a hardware technology park near Hyderabad
- * Up to 10 MGD of water can be immediately made available to industries in and around Hyderabad

POWER:

- * All towns and villages in the state are electrified
- * Three new private power projects sanctioned for A.P.
- * Installed generating capacity 5080 MW: Power generated in
1992-93 - 25,202 Million units
- * Planned additional capacity of 6762 MW of which 820 MW will be available in 1993-94
- 25% rebate in power bills (both demand and energy) is offered to eligible industries for a period of 3 years from the commencement of commercial production subject to a limit of Rs.50 lakhs in respect, of large and medium scale industries and Rs.30 lakhs for SSI

SINGLE WINDOW CLEARANCE:

The Andhra Pradesh Govt. has a Centralized Documentation and Clearances Centre (CDCC) for medium and large scale industries at the Commissionerate of Industries to ensure clearances. The Centre is the single window of the state for obtaining all clearances. A common application form of three parts covers all acts and laws under which clearances/permissions are required.

Stage I A: For obtaining Land use Certificate from the Director, Town and Country planning/Urban Development authority; Feasibility Certificate for power supply from APSEB; allotment of industrial space from APIIC.

Stage I B: For obtaining No Objection Certificate from AP Pollution Control Board.

Stage II: For obtaining clearances under Factory Act for building plans/machinery installation; clearances from Director, Town and Country Planning/Urban Development Authority, Health Department, Local body.

The CDCC (Centralised Documentation and Clearances Centre) closely monitors the implementation of each major and medium scale project through computerised data.

A State Level Empowered Committee sanctions project—specific infrastructural or other facilities.

In each district, there is a District Cell of the Centralised Documentation and Clearances Centre (CDCC), which operates at the Single Window for small-scale industries.

A District Industries Promotion Committee under the Chairmanship of the Collector performs a similar job for small scale industries.

Project management

1 Project management:

A traditional form of organization is not suitable for project management for the following reasons.

1. Project is non-routine, non-repetitive undertaking.
2. The relationships in a project setting are dynamic, temporary and flexible.
3. Project requires co-ordination of the efforts of persons drawn from different functional areas and contribution of external agencies.

Hence project management needs a different form of organization with sharper tools of planning and control and improved means of coping with human problems. Project management may be divided into

- Forms of project organization
- Project planning
- Project control
- Human aspects of project management
- Pre-requisites for successful project implementation

2 Forms of Project Organizations:

Project manager:

There is a need for identifying a person with responsibility for integrating the activities and functions of various departments and external organizations involved in the project work.

This person may be called as project manager or project coordinator.

Project organization may take one of the three forms.

1. Line and Staff organization:

- Primary responsibility of the person identified as project manager is coordinating the work of the people in the functional departments.
- Project manager acts in a staff position.
- He does not have the authority and responsibility of line management
- He serves as a focal point for receiving the project related information.
- It is a weak form of organization and not suitable for large projects.

2. Divisional organization:

- A separate division is set up to implement the project headed by a project manager.
- The project manager has full line authority.
- It facilitates the process of planning and control.
- Brings out the better integration of efforts.
- Strengthens the commitment for project related personnel to the objectives of the project.
- Improves the prospects of fulfilling time and budget targets.
 - It may suffer with inefficient use of resources.
 - It may result in unnecessary duplication of specialists.

3. Matrix organization:

The line and staff organization is conducive to an efficient use of resources but not suitable for an effective realization of objectives. The divisional form is suitable for effective realization of objectives, but is not suitable for efficient use of resources. Both the above are realized in the matrix form of organization., but at the cost of greater organizational complexity.

In the matrix organization, personnel working on the project have a responsibility to their functional heads as well as the project manager. The authority is shared between the PM and the functional manager.

The matrix organization is incongruent with the traditional organization theory. There is dual sub-ordination. It involves greater organizational complexity. Yet it seems to be a better vehicle for simultaneous pursuit of the twin objectives.

3 **Project planning:**

There will be a need for formal planning when the projects are larger in size and complexity.

Functions:

- It provides a basis for organizing the work on the project and allocating the responsibilities to the individuals.
- It is a means of communications and coordination among all those involved in the project.
- It induces people to look ahead
- It instills a sense of urgency and time consciousness.
- It establishes the basis for monitoring and controlling.

Areas of planning:

- Planning the project work
- Planning the manpower and organization
- Planning the money
- Planning the information system.
- Work break down structure

Logical and systematic break down of the project into component parts.

Project objectives and policies:

- Technical and performance objectives
- Time and cost goals
- Extent of sub-contracting
- Terms and conditions of contract.

4 **Project Life Cycle:**

1. Project development and preliminary engineering
2. Bidding and contract negotiation
3. Engineering design
4. Purchase and procurement
5. Construction
6. Commissioning

5 **Assessment of tax burden:**

Network diagrams

1.0 **Network diagram:**

Net work diagram is a mathematical model that uses small circles (nodes) connected by links (arrows) to represent precedence relationships.

Critical Path method (CPM) and Programme Evaluation and Review technique (PERT) are some of the network techniques.

Network diagrams are represented by 2 ways.

1. Activity on Arrow (AOA) - Arrows represent the activities.
2. Activity On Node (AON) - Nodes represent the activities. Arrows merely show the precedence relationships.

2.0 **Terms:**

Event :

It is a specific instant of time which marks the start or end of an activity.

It consumes no time.

Activity:

It is an operation or task in a project and consumes time and resources.

Critical activity:

The activity which, if consumes more than the estimated time, delays the project is called Critical activity. Those which do not delay the project are called Non-critical activities.

Dummy activity:

When two activities start at the same instant of time, the head events (Start event) are joined by a dotted arrow representing an activity called dummy activity. Dummy activity shows precedence relationship only and does not consume any time.

Critical path:

It is the longest path and consumes maximum time. Even if one activity on the critical path is delayed, the project time can not be met.

Earliest start Time (EST):

It is the earliest possible time at which an activity can start. It is calculated by moving from first event to the last.

Earliest finish Time (EFT):

It is the earliest possible time at which an activity can finish.

$EFT = EST + \text{Duration of the activity}$

Latest finish Time (LFT):

It is the latest possible time at which an activity can finish. It is calculated by moving from last event to the first.

Latest start Time (LST):

It is the latest possible time at which an activity can start.

$LST = LFT - \text{Duration of the activity}$

Float or Slack:

It is a spare time or a margin over and above the duration of a non-critical activity without affecting the project completion time.

Total float (TF):

It is the additional time which a non-critical activity can consume without increasing the project duration. It may affect the floats of the previous and subsequent activities.

$TF = LST - EST \quad \text{or} \quad LFT - EFT$

Free float (FF):

If all the critical activities start as early as possible, the surplus time is the free float.

$FF = EST \text{ of tail event} - EST \text{ of head event} - \text{Duration of the activity}$
(Preceding event is head event and Succeeding event is the tail event.)

Independent float (IF):

The use of independent float of an activity does not change the floats in other activities.

$IF = EST \text{ of tail event} - LFT \text{ of head event} - \text{Duration of the activity}$

3.0 PERT:

PERT can be employed where the project can not be defined easily in terms of time and resources required.

PERT utilizes the probability techniques in estimating the time required.

PERT is an event oriented system.

3.1 Time estimates:

1. Optimistic time (t_o): It is the shortest possible time in which an activity can be completed if every thing exceptionally well.
2. Most likely time (t_m): It is the time in which the activity is normally expected to complete under normal conditions.

3. Pessimistic time (t_p): It is the time in which an activity will take to complete if mostly the things go wrong. It is the longest of the three.
4. Expected time (t_e): $t_e = (t_o + 4t_m + t_p) / 6$
5. Standard deviation (S_t): $S_t = (t_p - t_o) / 6$
6. Variance = Square of the Standard deviation.
7. Probability that the project will meet the scheduled or due date is calculate from the equation:

$$Z = (D - T_e) / S_t$$

Where D = Due date or schedule date

Z = No. of Std. Deviations by which D Exceeds T_e .

T_e = Total project duration (Critical path)

Probabilities of schedule date for various values of Z are available in tables.

4.0 CPM:

It is a technique used for planning and controlling the most logical and economic sequence of operations for accomplishing a project. It is similar to PERT and is shown on network diagrams. It is an activity oriented system.

4.1 Steps:

1. Break the project into various activities systematically.
2. Label all activities.
3. Arrange all activities in logical sequence.
4. Construct the arrow diagram following the logic and sequence.
5. Number all nodes and activities.
6. Find the time required for each activity by previous experience or data.
7. Mark the times on the arrow diagram.
8. Calculate EST, LFT, LST and EFT.
9. Tabulate various times and mark EST & LFT on the arrow diagram.
10. Calculate the total float for each activity.
11. Identify the critical activities and mark the critical path on the arrow diagram.
12. Calculate the total project duration.
13. Crash the critical activities on the arrow diagram if project time is to be reduced.
14. Optimize the cost.
15. Update the network.
16. Smooth the network resources.

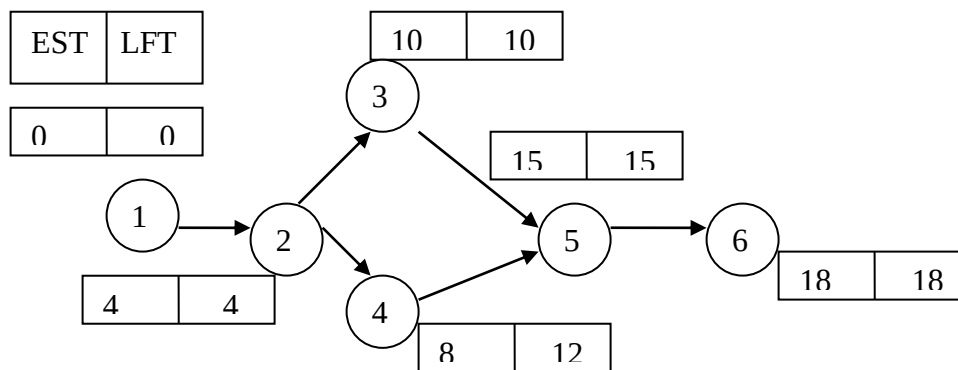
4.2 Differences between CPM & PERT:

	CPM	PERT
1	A deterministic model. Time is determined by past experience	A probabilistic model. Expected time is calculated from Optimistic, pessimistic and most likely times.
2	An activity oriented approach	An event oriented approach
3	Use of dummy activities is not necessary	Dummy activities may be needed
4	CPM marks critical activities	Basically does not demarcate between critical and non-critical activities.

4.3 Example:

Activity	Duration in days
A	4
B	6
C	5
D	4
E	3
F	3

Draw the network diagram and calculate EST, LST, EFT & LFT and floats. Mark the critical path.



Activity 1	Duration 2	EST 3	LST 4 (6-2)	EFT 5 (3+2)	LFT 6	Total Float 7 (4-3)	Free Float 8	Independent float 9
A	4	0	0	4	4	0	0	0
B	6	4	4	10	10	0	0	0
C	5	10	10	15	15	0	0	0
D	4	4	8	8	12	4	0	0
E	3	8	12	11	15	4	4	0
F	3	15	15	18	18	0	0	0

Unit-V

Behavioral aspects of entrepreneurs: Personality – determinants, attributes –and models. Leadership concepts and models. Values and attitudes. Motivation aspects. Change behavior. Time Management: Various approaches of time management, their strengths and weakness. The urgency addiction and time management matrix.

Personality:

It may be defined as dynamic organization within the individual of those psychological systems that determine unique adjustments to his environment.

It can also be defined as the sum total of ways in which the individual reacts and interacts with others. This can be described in terms of measurable personality traits that a person exhibits.

Personality determinants:

Personality may be considered to be made up of

- 1) Heredity 2) Environment 3) Situation

Heredity: Refers to these factors that were determined at conception.

- Physical stature
- Facial attractiveness.
- Sex
- Temperament
- Muscle composition and dexterity
- Energy levels
- Biological rhythms

All personality characteristics are not completely dictated by heredity. Some can be acquired or modified by other factors. Other wise they could be fixed at birth and cannot be altered by any amount of experience.

Personality Types (models):

Personality traits can be grouped to form personality modes. The specific characteristics can be grouped into categories. Four personality types or models can be thought of with properly grouped characteristics.

	High anxiety	Low anxiety
Extrovert	Tense, evitable, unstable Sociable, dependent	warm, composed, confident, trustable, adaptable, warm, sociable & dependent
Introvert	Tense, evitable, unstable, cold and shy	adaptable, Composed, confident, trustable, calm, cold and shy

Personality Traits:

1. Reserved - outgoing.
2. Less intelligent –More intelligent
7. Timid - Venture some
8. Trusting - Suspicious

- | | |
|--|----------------------------------|
| 3. Sensitive to feelings – Emotionally stable. | 9. Practical – Imaginative |
| 4. Submissive – Dominant | 10. Tough minded - Sensitive |
| 5. Serious – happy –go-lucky | 11. Self-assured – apprehensive |
| 6. Expedient – Lethargic | 12. Conservative – Experimenting |
| | 13. Uncontrolled – controlled |
| 14. Group dependent – Self-sufficient | |

Environment:

The environment, we are exposed to, plays a critical role in shaping our personalities.

- Culture in which we are living
- Our early conditioning
- Norms among our family friends and social groups.
- Influence of birth order.

The environment that a first-born child is exposed to is different from that of later born children.

Heredity sets the parameters or outer limits where as individuals fill potential is determined by the environmental influences.

Situation:

A third factor, situation further influences the effects of heredity and environment on personality. An individual personality while generally stable and consistent does change in different situations. Different demands in different situations call forth different aspects of our personality. Personality cannot be seen in isolation. Some situations impose certain constraints on behavior

Values:

Values represent basic connections that a specific mode of conduct or end state of existence is personally or socially preferable to an opposite or converse mode of conduct or end state of existence.

- Values carry individual ideas as to what are right, good or desirable values have both content and intensity attitudes.
Content attitudes say a mode of conduct or end-state of existence is important.
Intensity defines how important it is.
- Value system is a ranking of individual value according to their relative importance. It is the hierarchy of values such as freedom, pleasure, self-respect, honesty, obedience, and equality extra.

Importance of values:

- Values generally influence attitudes and behavior.
- Values affect job satisfaction or dissatisfaction.

Sources of our value system:

Culture	teachers
Parents	friends and others

Types of values	Levels of values
------------------------	-------------------------

- | | |
|---------------|----------|
| - Theoretical | Reactive |
|---------------|----------|

- | | |
|-------------|---------------|
| - Economic | Realistic |
| - Aesthetic | Manipulative |
| - Social | Socio-centric |
| - Political | Existential |
| - Religious | |

Attitudes

- Attitudes are evaluated statements either favorable or unfavorable concerning objects, people or events. They reflect that how one feels about something.
- Attitudes and values are different but they are closely related.
- Values are the broader and more varied in concept.
- Attitudes are more specific.
- Values contain a moral flavor of rightness desirability.
- Values people hold can explain their attitudes and behavior they engage.
- Attitudes shape behavior.

Sources of attitudes:

- Acquired from parents teachers and peer group members.
- The people whom are admire, respect or fear.
- Advertisement manages many other attitudes.

Types of attitudes:

- 1) Job satisfaction:
 - General attitudes towards his job.
 - A high level of job satisfaction holds positive attitudes towards job.
 - Dissatisfaction leads to negative attitudes.
- 2) Job involvement:
 - Degree to which a person identifies psychologically with his job and considers his performance level important to his job.
 - Fewer absences and lower turn over.
- 3) Organizational behavior:
 - A state in which an employee identifies with a particular organization and the goals.
 - A better indicator of turn over.
 - Loyalty towards organization.

Theories of attitudes:

- 1) Cognitive dissonance theory:
 - Any incompatibility between two or more attitudes or between attitudes and behavior.
 - A state of minimum dissonance as it can't be avoided totally.
 - Rewards influence the degree to which individuals are motivated to induce dissonance.
 - Greater the dissonance the greater the pressure to reduce it.
- 2) Self perception theory:
 - Attitudes are used after the fact to make sense out of the action that has already occurred rather than devices that precede and guide action.

- We are very good at finding reasons for what we do but not so well at doing what we find reason for.
- 3) Attitude surveys:
 - The information about the attitudes of the employees can be obtained through attitude surveys.

Attributes: Influencing organizational behavior

1. Locus of control: The degree to which people behave. They are masters of their own fate.
 - Internals – behave that they control their destinations.
 - Externals – behave that others control their lives.
2. Achievement orientation
3. Authoritarianism: -status& power.
4. Machiavellians:
 - Degree to which an individual pragmatic, maintains emotional distance and delivers that end can justify means.
5. Self Esteem
6. Self Monitoring
7. Risk taking

Behavioral aspects of Entrepreneurs:

Attributes:

- Positive feed back from good job.
- Initiate confidence in employees.
- Be fair and objective, not vindictive.
- Practice what is preached.
- Have down to earth, person-to-person approach.
- Act as a part of a team.
- Be compassionate and understanding.
- Create an atmosphere of acceptance and trust.
- Listen attentively.
- Clear priorities.
- Be available for discussion.
- Communicate effectively.
- Provide feed back for good and bad work.
- Talk openly and give straight answers about what is happening.
- Encourage advancement.
- Give direction to the work unit.
- Be proud and supportive to employees.
- Have a sense of vision.
- Project a positive attitude.
- Set good example, friendly, not moody keeps promises, planes through red tape.
- Willingness to assume risks

Leadership

1.0 Leadership:

It is a process of influencing and supporting others to work enthusiastically towards achieving the objectives.

1.1 Features of leadership:

1. Leadership is a continuous process of behavior – not one-shot activity
2. It can be seen in terms of relationship between a leader and his followers (Individuals or groups) which arises out of their functioning for common goals.
3. The leader tries to influence the behavior of individuals or groups of individuals around him to achieve common goals.
4. The followers work willingly and enthusiastically to achieve these goals. Thus there is no coercive force which induces the followers to work.
5. Leadership gives an experience of help to followers to attain common goals.
6. Leader feels the importance of individuals and gives them recognition and conveys them the importance of the job they perform.
7. Leadership is exercised in a particular situation, at a given time and under specific set of circumstances. It implies that the leadership styles are different under different situations.

1.2 Difference between leadership and management:

Factors	Leadership	Management
Source of power	Personal abilities	Authority delegated
Focus	Vision & purpose	Operating results
Approach	Transformational	Transactional
Process	Inspiration	Control
Emphasis	Collectively	Individualism
Futurity	Proactive	Reactive
Type	Formal & informal	Formal

1.3 Importance of leadership:

1. Motivating employees
2. Creating confidence
3. Building morale

1.4 Leadership theories:

1. Charismatic leadership theory
 - A leader is born and is not made
 - A leader has charisma (gift) which acts as influencer

- The charismatic leader has high level of self confidence, dominance, and strong conviction in his beliefs.
 - Charismatic leaders communicate a vision or high level goal that captures the commitment and energy of his followers.
 - The qualities are inborn and can not be enhanced through education
 - These inborn qualities are sufficient for a leader to be successful
2. Trait theory
- Leadership traits are not completely inborn but can also be acquired through learning and experience.
 - The traits in a successful leader are physical and constitutional factors. Intelligence, self confidence, will, dominance, and surgency (talkative, cheerfulness, enthusiasm)
 - Innate qualities: 1. Physical features 2. intelligence
 - Acquirable qualities: 1. Emotional stability 2. Human relations 3. Empathy 4. Objectivity 5. Motivating skills 6. technical skills 7. Communicative skills 8. social skills
3. Behavioral theory
- Strong leadership is the result of effective role behavior
 - It is shown by a person's acts more than by his traits
 - Task-related functions and group maintenance functions are the two tasks to be fulfilled by a leader
 - Leadership behavior prescribed various leadership styles
4. Situational theory
- Situation in which leadership is exercised also plays an important role.
 - The effectiveness of the leadership depends on two factors- 1. leader's behavior 2. situational factors
 - Leader's behavior:
 1. Leader's characteristics
 2. Hierarchical position
 - Situational factors:
 1. Subordinate's characteristics
 2. Leader's situation
 3. Group factors
 4. Organizational factors

2.0 Leadership Styles:

Leadership styles are the patterns of behavior which a leader adopts in influencing the behavior of his followers. These styles are based on behavioral approach or situational approach.

1. Based on behavioral approach:
 1. Power orientation
 - Autocratic leadership
 - Participative leadership
 - Free-rein leadership
 2. Leadership as a continuum

- A variety of styles of leadership between two extremes of autocratic and free-rein.
- A broad range of styles on continuum moving from authoritarian leadership behavior at one end to free-rein behavior at the other end.
- Use of authority by the manager gradually decreases as the style moves from one end to the other while the freedom of subordinates increases.

3. Employee-production orientation

- Leadership depends on two dimensions 1. Initiating structure 2. Consideration.
- Leadership behavior is plotted on separate axes rather than a single continuum.
- The 4 quadrants show various combinations of initiating structure and consideration.

High consideration & Low structure	High consideration & High structure
Low structure & Low consideration	Low consideration & High structure

TIME MANAGEMENT

An abundantly available natural resource which is equally available to each and every one without any discretion what so ever, which can not be saved for future use, which can not be regained once lapsed, which continuously gets consumed whether used or not is the time.

Approaches of time management:

1. The “get organized” approach: (order)

- Assume that most time management problems are due to chaos - Lack of order.
- Things are constantly falling through the cracks
- The systems are impossible for the problems
 - Organization of things→ creating order for every thing
 - Organization of tasks→giving order & sequence to jobs.
 - Organization of people→ Defining roles, delegating, creating tracking systems.

This approach goes beyond personal application into organizational practice.

Strengths:

- Saves time and leads to greater efficiency.
- Economizes the efforts
- Brings clarity and order.

Weaknesses:

- It becomes an end rather than a means, to greater ends.
- More time is spent in organizing rather than producing.
- Danger of over structured leading to inflexible and mechanical.

2. The warrior approach: (Survival and independent production)

- Focus on the production of personal time to focus and produce.
- A situation where there is more to do than the staff can handle.
- Something has to be done otherwise system becomes an avalanche that will burry us alive.
- Defending approach
- To focus on high leverage Independent action
 - Insulation (creating protection through secretaries, closed doors etc.)
 - Isolation (Aloneness creates uninterrupted time.)
 - Delegation (assigning tasks to others to find time for more high leverage tasks.)

Strengths:

- Assuming personal responsibility

- Have quiet and uninterrupted time to do high leverage tasks
- Useful for highly creative works

Weaknesses:

- This approach assumes that others are the enemy.
- It creates self-fulfilling prophecy.
- People sense that they are being put off and they fight back
- The approach may not be effective in an interdependent reality.

3. The goal approach (achievement):

- Focuses on “know what you want and focus your effort to achieve it”.
- Includes techniques such as long term, midrange, and short term planning
- Goal setting, visualization, self motivation
- Creating positive mental attitude

Strengths:

- Approach leads to good performance and achievement
- Marshal the forces
- Focus on energy
- Refuse to be distracted
- Setting goals leads to accomplishing.

Weaknesses:

- Focused drive and single mindedness to achieve goals can blind people to imbalances in their life
- They may have glamorous public life but an empty private life.
- Lack in their relationship satisfaction deep inner sense of integrity.

4. The ABC approach: (prioritization and values identification):

- It builds on the goal approach and adds the important concept of sequence
- Concentrates on most important tasks first
- Values clarification and task ranking

Strengths:

- First things first – gives order and sequence
- Differentiation between the tasks and encouragement on focusing top priority tasks.
- The deeper analysis around values is helpful and productive.

Weaknesses:

- Values classification doesn't recognize that there are principles, natural laws that govern quality of life.
- May lead to frustration and failures
- Priorities can change when they are fulfilled.

5. Magic approach: (Technology):

- Based on the assumption that a right tool will give us the power to create quality in our life.
- Tools may be the right calendar, right planner, right computer program etc.
- The tools typically help us keep tracking priorities, organize tasks, and more easily access key information.
- Systems and structures help make us more effective

Strengths:

- Tools help in keeping track of priorities, reminding goals, organizing the tasks and easy accessing of the frequently used information.
- Tools are symbols of hope.

Weaknesses:

- Even a best tool is no substitute for vision, judgment, creativity, character or competence.
- A tool can help us in creating but can not create
- For many people tools seem rigidly structured and unnatural.
- Very few use the tools in the way they are designed

6. The time management 101 approach: (skills):

- This approach is based on the paradigm that the time management is essentially a skill.
- In order to function effectively we must know to master certain basics like
 - Using a planner or appointment calendar
 - Creating “to do” lists
 - Setting goals, delegating, organizing and prioritizing
- It is an organizational approach
- Social literacy is required for survival.
- As part of human resource development programs many firms make tapes, booklets, and courses designed to teach the basics

Strengths:

- Skills are improved through training.
- Some improvements are made in terms of work related skills.

Weaknesses:

- The depth and quality of training is the primary issue
- Are they inline with the principles?
- Do they propagate inaccurate assumptions about the nature of life & effectiveness?
- Skills alone do not provide the answer to the solutions
- More than skills and techniques, individual and organizational quality is a function of aligning both personal character and personal behavior with principles

7. “The go with the flow” approach: (harmony and natural rhythm):

- Emphasis is placed on the congruity of inner self and once harmony with the flow of nature
- All living beings have certain vibrations and natural body rhythms. The modern style of living and the tools used are working against the natural rhythm, creating serious illness and problems.
- Learning to go with the flow and getting back to the natural rhythm of living will help us in achieving spontaneity and serendipity which is natural to our being.

Strengths:

- Sensitizes us to the values of timeless moments and creates more of them in our lives.
- Moves us away from the dominance of “urgent”.
- It creates and encourages internal and external harmony.

Weaknesses:

- Vital elements such as vision, purpose, and balance are frequently missing.
- It is reaction to urgency addiction, an escape rather than aid to creating quality of life.

8. The recovery approach (self awareness):

- Influenced by an early role model of family culture and individual will become a “perfectionist”.
- Afraid to delegate
- Tends to micro manage
- Spends an inordinate amount of time on the projects beyond the effective utilization of the resources.
- Over committed and over worked out of fear of rejection.

Strengths:

- Helps identify nature and sources of dysfunctional time management habits.
- Leads to greater self awareness and prepare people to make fundamental changes and improvements

Weaknesses:

- The suggested methods for recovery are very much varied.
- Does not provide for confident response to higher priorities?
- Does not recognize extrinsic realities that govern quality of life.
- Priority is often defined by urgency, circumstances or other people.

Urgency addiction:

The primary functions that drive our choices concerning the use of time are (1) Urgency (2) Importance. One of the factors will be prominent through which we view our time and our lives.

The urgency addiction Experience:

- Creates predictable, reliable sensations.
- Becomes the primary focus and absorbs the attention.
- Temporarily eradicates pain and other negative sensations.
- Provides artificial sense of self worth power, control security, intimacy, accomplishment
- Exacerbates the problems and feelings it is sought to remedy.
- Worsens functioning, creates loss of relationship.

Urgency addiction is a self-destructive behavior that temporarily fills the void created by unmet needs. the tools and approaches of time management often feed the addiction.

Urgency feel:

- Stressful
- Pressed
- Tense
- Exhausting

It is some times

- Useful
- Exhilarating
- Successful
- Validated

Importance --- words used in expressing;

Confident

Fulfilled

Peaceful

Meaningful

On track

Urgency and importance are two factors in varying degrees in time management matrix. The time managing matrix categorizes all our activities into 4 quadrants.

The time management matrix:

The time management matrix categorizes all an activities in to 4 quadrants